

FINITE AND INFINITE MATHEMATICS

Random Bits, Finite Catalogs, and Arbitrary Cuts of Arithmetic

Exact noise thresholds, Boolean universality,
and the strength of induction

Research report prepared for Vladimir Reshetnikov

Developed and written with ChatGPT

5 October 2026

Central question. How much random information can a mathematical universe encode, and how does the answer change when a finite catalog is replaced by all of the codes in a countable nonstandard model?

Main result. One arithmetic formula has an exact almost-sure classification for every independent Bernoulli law. Suitable biases realize every successor cut, and even every monotone system of cuts assigned to the finite conjunctions of countably many independent predicates.

Status. Complete mathematical proofs and finite computational checks are supplied. The fair-coin standard-cut result is credited to Elliot Glazer; the source-coding foundations are classical. The additional classifications are proposed research contributions, with no claim of established priority or of a machine-checked formalization.

Abstract

We formulate a coding problem with complementary finite and infinite forms. For finitely many random bits, the resource is the number of candidate words allowed in a catalog. For a countable model M of Peano arithmetic, the resource is the family of binary strings internally coded by its elements. The fixed formula asserting that a predicate is coded below a defines a successor-closed initial segment, its coding cut.

For arbitrary pairwise-independent bits with biases p_t , let b_t be a most likely bit and $\varepsilon_t = \min(p_t, 1 - p_t)$. We prove that the random coding cut is almost surely

$$\text{Cut}(b) \cap \{a \in M : \sum_{t < a} \varepsilon_t < \infty\}.$$

The sums are external countable sums. Every cut is realized by a full-support, nonatomic product law, even after prescribing any finite collection of coordinate probabilities strictly between zero and one. More strongly, any inclusion-monotone assignment of cuts to the nonempty finite subsets of $\mathbb{N}$ is realized as the coding cuts of the corresponding conjunctions of independent predicates. The biases may all tend to zero. A classification of all finitary Boolean observations follows from their minimal supports that change the all-zero output.

We also establish a three-regime induction theorem, a sharp distinction between marginal fairness and pairwise independence, category and finite-limit obstructions, and stability under locally summable perturbations. The finite problem has an exact optimal-list formula, a summability criterion for uniformly bounded budgets, the classical normal source-coding transition, and an explicit sparse Poisson staircase. The report closes with a source and proof audit, a concrete ProveIt formalization plan, and fourteen further research problems.

[write] **Status and trust boundary, added 6 October 2026, batch 114.** This manuscript is filed in ProveIt as the single-source research report `random-bits-arithmetic-cuts` (provenance, the sources as the write read them, relation to the repository with the formal status of each statement, reading conventions and the collected non-claims in Section 1.3). It is AI-assisted (its title page says it was developed and written with ChatGPT), unrefereed, and *not formalized*: no Lean or Rocq declaration exists for any theorem, lemma, proposition or corollary of this report. The one sentence of the text with a formal counterpart in the repository is the passing remark of Section 1 that PA has no finite models (`no_finite_PA_model`, `model_carrier_not_finite`). *What rests on what.* The infinite-model results rest on the written proofs alone. The finite computations of Section 12 check small finite cases and regenerate the plotted data; they verify no infinite theorem. The fair-coin standard cut (Corollary 3.2) is credited to Glazer’s 2023 announcement, and the finite source-coding asymptotics and the pairwise atom bound to the classical literature. Section 11 is the source’s *proposal* for a formalization, not a description of the repository. The write adds Remark 13.1, with a proof, and Section 13.1, which carries the fourteen questions of Section 13 and the source’s priority statement as open questions with their status.

Contents

1	The problem and its mathematical setting	4
1.1	Why this connects Glazer and ProveIt	4
1.2	What is established here, and what is inherited	5
1.3	Provenance, sources and reading conventions	5
2	Arithmetic coding and the deterministic cut	9
2.1	Internal and external conventions	9

3	The exact classification for independent noise	10
3.1	Three elementary examples	12
4	Realizing any cut without changing prescribed finite data	12
5	Independent predicates and complete Boolean coding spectra	13
5.1	A useful one-sided form of the product classification	13
5.2	Universality for all finite conjunctions	14
5.3	Every finitary Boolean observable	16
5.4	Examples and sharp operation laws	18
5.5	A series threshold for repeated independent thinning	19
5.6	Universality with vanishing probabilities	19
6	An exact induction phase diagram	20
6.1	Keeping track of the language	20
6.2	Bounded induction is equivalent to local coding	21
6.3	An infinite locally finite predicate defines the standard cut	22
6.4	Three regimes for independent sparse predicates	23
7	Category, finite observations, and two orders of limits	24
7.1	A finite observation never certifies noncoding	25
7.2	Stability under locally summable perturbations	26
8	How much independence is actually necessary?	27
8.1	Limits of the pairwise extension	28
8.2	A general dependent-law criterion	29
9	The finite candidate-budget problem	29
9.1	Exact optimization and the infinite limit	29
9.2	An exact Hamming-layer formula	31
9.3	The fixed-bias critical window	32
9.4	A sparse-noise Poisson staircase	33
9.5	Why finite observations alone cannot certify noncoding	33
10	Consequences linking the constructions	35
10.1	What the finite probabilities say about internal codes	35
11	A proposed formalization route in ProveIt	36
11.1	Use raw first-order arithmetic semantics	36
11.2	Suggested proof modules	36

12 Verification and source audit	37
12.1 Mathematical checks	37
12.2 Executed finite checks	38
12.3 Source boundary	38
13 Further research questions and topics	38
13.1 Further questions and research	40

1 The problem and its mathematical setting

A finite binary word can always be encoded by an ordinary natural number. That observation changes character inside a nonstandard model of arithmetic. An interval that the model regards as finite can contain infinitely many elements externally. There are only countably many internal codes when the model is countable, but continuum many external assignments of bits to such an interval. Randomness makes this difference visible to a single formula.

The finite question is a resource question: how many candidate descriptions must one allow before a random word is likely to have a description on the list? The infinite question is a definability question: at which endpoints does the entire random initial segment have an internal description? This report treats these as two parts of one problem, while keeping the change in quantifiers explicit.

- Problem 1.1** (Finite catalogs and infinite coding cuts). (a) Given independent bits on N sites and a catalog of at most K words, determine the largest possible probability of finding the observed word in the catalog. Determine the budget needed at a fixed confidence, including regimes in which the individual biases approach zero.
- (b) Given a countable nonstandard model $M \models \text{PA}$ and an external random predicate $P \subseteq M$, classify the endpoints a for which $P \cap [0, a)$ is internally binary-coded.
- (c) For independent predicates whose coordinate biases are uniformly bounded away from one, determine which cuts, and which systems of cuts under Boolean operations, can occur. Determine whether finite observations, topological genericity, or induction principles constrain the answer.

There are no finite models of PA : its successor is injective and omits zero. Part (a) is consequently a finite coding problem, not a purported finite model of arithmetic. Nor is a catalog of K externally selected codes the same as the model's unrestricted collection of codes.

[write] **6 October 2026.** The first sentence of this paragraph is formalized in the repository (`no_finite_PA_model` and `model_carrier_not_finite`, by the argument given here; details in Section 1.3). Nothing else in this report is.

1.1 Why this connects Glazer and ProveIt

Glazer's October 2023 MOPA talk announced that one formula defines the standard cut almost surely under fair independent bits in every countable model of PA , and even of Robinson arithmetic Q [1]. Our binary-code proof reconstructs the PA case; the public abstract does not specify his formula. The refinements pursued here concern nonuniform noise, prescribed cuts, Boolean operations, and finite information budgets. They give a direct mathematical reason to expect interest, without assuming a personal endorsement or a current professional role.

The relevant ProveIt connection is its existing arithmetic infrastructure. At the inspected repository snapshot, its list-coding development gives genuine arithmetic formulas for finite lists and base digits, and its non-finite-axiomatizability development includes external finite beta coding in arbitrary raw PA models and a definable-cut induction argument [7, 8, 9, 10]. These are concrete interfaces for future formalization. They are not a formal verification of the present probability theorems.

1.2 What is established here, and what is inherited

The uniform fair-coin conclusion is prior work. Selecting the most probable words in a finite source and the fixed-bias normal approximation belong to classical lossless compression [3, 4]. The finite moment method used for pairwise-independent bits is likewise standard extremal probability. Full proofs are included because their combination explains the finite/infinite boundary.

The proposed contributions are the exact arbitrary-bias coding-cut formula, arbitrary-cut and countable Boolean-spectrum realization, the finite-marginal and weak-convergence separation, and the resulting induction classification for zero-modal product laws. We did not find these precise formulations in the sources inspected. This is a bounded literature assessment, not a priority certificate. The report does not assert that a named published open problem has been settled or that the results constitute an established breakthrough.

Result	Role in the report
Fair independent bits reveal the standard cut	Prior result; a PA proof is reconstructed as a corollary.
Arbitrary product-law formula	Complete classification proved here.
Countable conjunction spectra	Monotonicity is the exact realization condition under the stated independence and bias bounds.
Induction phase diagram	Exact almost-sure classification for zero-modal independent noise in a nonstandard model.
Finite optimal catalogs and normal window	Classical tools, proved and connected to the infinite theorem.
Sparse polynomial budget staircase	An explicit consequence of optimal ranking and the binomial-to-Poisson limit.
Formal verification	A proposed plan; only the stated finite calculations were executed for this report.

1.3 Provenance, sources and reading conventions

[write] **Added 6 October 2026, batch 114.** *Provenance.* This report was built from one manuscript, the archive `Random_Bits_Arithmetic_Cuts.zip` (817,508 bytes, wrapper directory `random_bits_arithmetic_cuts/`, 17 files), which arrived in ProverIt’s `docs/incoming` in commit 2399df2bd (5 October 2026; not part of a session bundle) and was placed as batch 114 in 99053b5d1. The text is the delivered `random_bits_arithmetic_cuts.tex` (2,647 lines, 37-page PDF), shipped as `article.tex`. Its title page reads “Research report prepared for Vladimir Reshetnikov” and “Developed and written with ChatGPT”; its PDF author field reads “Research report prepared with ChatGPT for Vladimir Reshetnikov”. Its repository pin is 3b9458b31f459b480ab43a068d7cc56f4cfd9bb28, the placement commit of batch 98D (Section 12 and `SOURCE_AUDIT.txt`). The delivered PDF and `SHA256SUMS.txt` (16 entries, all verified at placement and again at the write) are not shipped; both are retrievable from the arrival commit. This is a single-source report, so no merge choice was made. The write prefixed the 87 delivered labels with `rbc:` (the 22 labels `bs:...` became `rbc:bs:...`) and updated every reference. It added this subsection, the notes marked [write], Remark 13.1, Section 13.1, a label and the reference format `Q1, ..., Q14` on the fourteen questions of Section 13, and it changed the title of Section 11 (the delivered title is kept in the note there). No statement, proof or number of the source was changed: every theorem, equation, figure and section keeps its delivered number.

The sources, as the write read them (5–6 October 2026).

- Glazer [1]: the official seminar abstract. It announces that the standard cut of a countable model of PA, “or even Q”, is uniformly definable relative to a randomly chosen predicate; that for a countably

saturated model no formula yields a set measured by the completed product measure on 2^M ; and a programme of combinatorial ideals on 2^M for models of any cardinality. It names no formula. Sections 1 and 3 describe it accurately.

- Abdul-Quader and Schmerl [2], arXiv version 3: all models there are models of PA, “definable” means definable with parameters, a *class* is a set whose restriction below each element is definable, and Theorem 19 states that in a nonstandard model no class is strongly CP-generic. This is the obstruction invoked in Question Q5; Remark 13.1 spells it out.
- Benjamini, Gurel-Gurevich and Peled [5], arXiv:1201.3261v1 (16 January 2012; its comments say the manuscript had been on the authors’ websites since 2007): Proposition 25, formula (12), is the maximal probability that all n bits equal one under pairwise independence with marginal p , obtained by adapting the closed forms of Boros and Prékopa. At $p = 1/2$ it gives $1/(n+1)$ for odd n and $1/(n+2)$ for even n , as Theorem 8.2 says (after complementing coordinates, every word is the all-ones word). Boros and Prékopa [6] were read neither by the source nor by the write.
- Kontoyiannis and Verdú [3], Chen, Effros and Kostina [4]: the arXiv records were located (arXiv:1212.2668 and arXiv:1902.03366; the DOI of the second is confirmed there); their theorems were not read by the write. The source proves Theorem 9.5 itself and credits the expansion to them; nothing in the report depends on the exact form of their statements.
- The four repository files the source read [7, 8, 9, 10] are byte-identical at the pin and at the write’s HEAD 3988bf5c4; their relation to this report is described below.

What was checked at intake. The batch-114 dossier read Sections 1–11 in full and checked by hand the Borel–Cantelli tail with finite-change invariance, the nullity of fixed patterns with the countability of codes, the diagonal marker recursion and its marker property (5.3), the product-weight error bound in (5.5), the second-moment bound (8.2), the two moment polynomials of Theorem 8.2, the induction equivalence and the locally finite Σ_1 list formula. It found no error. The write reread the whole text, including Sections 12 and 13 and the bibliography, and found none either. Independent intake programs confirmed the Hamming-layer formula of Theorem 9.4 against sorted atom sums for $N \leq 8$ at $p = 1/10$ and $p = 1/3$, the three values of the table in Section 10 to all printed digits, and the first two moments and the atoms of the pairwise extremizers for $N \leq 10$. The delivered script was rerun on copies at placement and at the write (note in Section 12).

Relation to the repository; formal status. Placement in the collection confers no formal status. At HEAD the repository’s formal developments contain the following, and nothing else that bears on this report apart from the standalone copies named at the end of the first item. (Wording corrected after the independent check of 6 October 2026; the first wording ended “and nothing else that bears on this report”).

- *Formalized: the remark that PA has no finite models* (Section 1). Lean: `no_finite_PA_model` and `model_carrier_not_finite`, with the model-independent lemma `finite_carrier_cannot_have_injective_successor_missing_zero`, in `Logic/PeanoArithmetic/NoFiniteModel/Lean/NoFiniteModel/Theorem.lean`; Coq counterparts (for example `peano_arithmetic_has_no_finite_model`) in `Logic/PeanoArithmetic/NoFiniteModel/Coq/NoFiniteModel.v`. The Lean theorems are stated for the shallow interface `SetTheory.PA.Model` of `Logic/Interpretability/PAHF/Lean/PAHF/PASyntax.lean`, whose induction field ranges over every Lean predicate, the kind of interface that Section 11 says cannot express nonstandard models. The proof uses only that successor is injective and omits zero, exactly as the remark does, so the argument covers every first-order model of PA; the formal statement is the one for the shallow interface. [Added after the independent check of 6 October 2026.] Same-named copies of the three Lean declarations, in the namespace `ProveIt.Optimizations.NoFiniteModel` and for a structure `ShallowPAModel` with only zero, successor and the two successor axioms (no induction field), are in `Optimizations/Logic_Foundations_Optimizations.lean` (commit 88e56dc38). That file is not imported by the root Lean library (`lakefile.toml`, `ProveIt.lean`).
- *Ingredients only, not statements of this report.* `finite_list_beta_code` and `finite_vector_beta_code`, in `Logic/PeanoArithmetic/NotFinitelyAxiomatizable/Lean/PAFi`

`niteBasisReduction/FiniteBetaCoding.lean`: in every raw arithmetic structure satisfying all axioms of PA (`RawPASatisfies`), an externally finite list of arbitrary, possibly nonstandard, elements (a Lean list, or a vector indexed by `Finn`) has a beta code whose entry at the numeral i is the remainder modulo $1 + (i + 1) \cdot \text{step}$, the convention of (6.4). That is the step “encode this externally finite list” in the proof of Lemma 6.5 (for standard n). It is not Lemma 6.1, whose sequences have a possibly nonstandard internal length x and need the internal Chinese remainder theorem of PA; the binary relation `Bit` of (2.1) is not formalized at all. `not_sat_inductionForm`, in `.../PAFiniteBasisReduction/EvaluatorCutContract.lean`: under that module’s evaluator `Contract`, the induction instance of the specific formula `cutAt evalFormula` is false, because that formula defines exactly the standard elements (`sat_cutAt_iff_isStandard`; the sealed form is `not_sat_sealed_induction`). This is the pattern of Corollary 6.2 and Lemma 6.5 (a formula that defines a proper successor-closed set containing 0 refutes its own induction instance), but for a different formula and with no predicate P ; it says nothing about χ or ℓ . `raw_above_of_nonstandard` (in `FiniteBetaCoding.lean`): a nonstandard element lies above every numeral, the fact behind “every standard index i would satisfy $i < n$ ” in the same proof.

- *Not formalized*: every theorem, lemma, proposition and corollary of this report; nothing probabilistic, no coding cut, no external summability cut, no `Bit` coding and no statement about induction in (M, P) . The source’s plan (Section 11) is a proposal, carried as Question Q13.

Stale claims. The source’s statements about the repository (Sections 1, 11 and 12) were checked against HEAD and are accurate: the `ListCoding` README calls its formula specifications metatheorems about the standard interpretation; `NotFinitelyAxiomatizable` works with raw first-order structures, not with the meta-level `PA.Model`; the two Lean files contain what is said of them. The source did not cite the `NoFiniteModel` project, present at its pin since July 2026, which formalizes its remark on finite models. Its statement that searches found no repository treatment of random predicates or coin flipping is still true: Glazer’s other work is cited elsewhere (surreal-number reports, and `measurable-box-games` and `naming-elementary-embeddings` in this collection), but no other file cites the coin-flipping talk or treats random predicates on models of arithmetic.

Neighbouring reports (under `SetTheory/Cardinals/docs/reports/ordinals-and-order-types/`): `noisy-parity-cubes` (same batch; *see also*): its Proposition `np:found:biased` makes a total parity label measurable for the completed product measure of independent `Bernoulli( $q_i$ )` coordinates exactly when $\sum_i \min(q_i, 1 - q_i) < \infty$, the minority-mass condition of Theorem 3.1 and Theorem 9.2, by the same first Borel–Cantelli step (almost every point is a finite modification of the modal sequence); the theorems differ and neither source cites the other. `robust-neutral-choice` (same batch) has a different biased threshold, $\sum_i (p_i - 1/2)^2 = \infty$. `measurable-box-games` answers Glazer’s *choiceless box game paradox* (arXiv:2211.10474) and uses product measures on random bits, a different problem with no shared theorem; the other new reports of batch 114 (`freiling-symmetry-blacklists`, `bounded-width-power-set-compression`) share a delivery template, but no theorem and no citation.

Reading conventions. No symbol was renamed. The source reuses the following letters; read them by section.

Symbol	Main meaning	Other meanings; tempting false readings
E (roman)	exponentiation $t \mapsto 2^t$, named in a definitional expansion (§§2, 6)	not expectation, which is $\mathbb{E}$ (§8); E_S is the error term of (5.5)
$\mathbb{P}$, Pr	probability (§§5–8, resp. §9)	the same notion
$S(w)$	the external summability cut (3.1)	not a sum computed inside M ; S is also a finite index set $S \subseteq \mathbb{N}$ (§5) and the successor symbol (proof of Theorem 6.3)
$\mathcal{D}_a = \mathcal{D}_a$	the countable family of internally coded subsets of $[0, a)$	$\mathcal{D}(f)$: minimal sensitive supports (5.6); $\mathcal{A}_x$ (§9) is $\mathcal{D}_x$ again; D_f , D : deviation predicates
A_a	the event $a \in \text{Cut}(P)$ (§7)	A_S : marker sets (§5); A, B, C : marker sequences in (5.12); A_t : fair bits and A : a predicate (§8, Remark 5.3)
θ_P, ϑ	the same coding formula, (2.2) and (6.1)	$\theta \in (0, 1]$: the captured fraction of a layer (Theorem 9.4)
χ, ℓ, Λ	bare-language $\Sigma_1(P)$ formulas (6.5), (6.9) and the bounded matrix (6.8)	χ, ℓ are not bounded; λ is the Poisson parameter
δ	a bias bound ($\delta < 1/2$ in Theorem 5.2, $\delta < 1$ in Theorem 5.5)	$\delta(z, P, \bar{a})$: a bounded formula (Theorem 6.3); δ_n : conditional bounds (§8); a small $\delta > 0$ in proofs
$\varepsilon_t, \varepsilon_i$	minority mass $\min(p, 1 - p)$ (3.2)	$\varepsilon > 0$: a threshold (Proposition 5.8)
c	an internal binary code	also constants ($c = \delta/2$ in (5.4), $c \leq 1/4$ in §3)
W	$W_w(a)$ (3.1); W, W_x external sums (Theorem 6.6)	the number of ones (Theorem 8.2); $W_N \sim \text{Bin}(N, p)$
q	prescribed marginals q_t (Theorem 4.2)	$q(x)$ (Lemma 5.1); mismatch probabilities q_t (Theorem 8.1); the atom bound q_N ; a modulus $q \geq 6$; a quotient in (2.1)
Q	$Q_N(K)$, the optimal catalog mass	Robinson arithmetic Q ; a predicate Q (Propositions 2.3, 7.5)
K	a catalog size	a cut K in (5.12); a catalog of K words is not the model's family of codes (§9)
b, B	the modal predicate b_t (3.2)	$b_f = f(\mathbf{0})$; an element b in proofs; $B_N(r)$: partial binomial sums; a predicate B
a	an endpoint in M	$N = 2a - 1$ or $2a$ (Theorem 8.2); $a_N = \prod(1 - \varepsilon_i)$; a_j : constraint endpoints
I	a cut	$\mathsf{I}\Delta_0, \mathsf{I}\Sigma_1$ (sans serif): induction schemes
cut	nonempty, downward and successor closed; M allowed	no closure under $+$ or $\cdot$ is assumed; a cut need not be definable
finite	externally finite (finite in the metatheory)	an interval $[0, a)$ with nonstandard a is internally finite but externally infinite
$\text{Cut}(P) = M$	every bounded restriction of P is coded (P is a class)	not that P itself is coded or definable

What the source does not claim (from the title page, the text, the delivered README and the audit files; all kept in force). The fair-coin standard cut is Glazer's: the PA case is reconstructed with the source's own formula, and his Q case and the uncountable and measurability problems are not claimed. Optimal finite ranking, the normal source-coding transition and the pairwise atom bound are classical; the sparse Poisson

staircase is derived, not presented as a new source-coding theorem. No established priority and no solution of a named open problem are claimed; the literature and repository searches were focused, not exhaustive. The finite checks neither simulate a nonstandard model, nor verify an infinite theorem, nor establish priority; the reviews were an internal audit, not peer review; no Lean or Rocq formalization was made. The Π_3^0 bound is not claimed optimal; the catalog bound Kq_N is not claimed sharp for $K > 1$; Theorem 4.2 keeps finitely many prescribed marginals but does not match arbitrary correlated data; Theorem 9.2 concerns prefixes of one infinite bias sequence, not the triangular array $p_N = \lambda/N$; the induction theorems need a nonstandard model (the standard-model exception); no failure of bare $\Delta_0(P)$ induction is inferred by deleting E; the countable conjunction theorem gives no law for countably infinite conjunctions or disjunctions (Remark 5.4); the vanishing condition of Proposition 5.8 is not vanishing in the internal order; nothing is claimed about uncountable or countably saturated models; no personal endorsement by Glazer is assumed. The write adds that none of its own additions is formalized either.

2 Arithmetic coding and the deterministic cut

2.1 Internal and external conventions

Fix a countable model $M \models \text{PA}$. Except where explicitly mentioned, M is nonstandard. Its standard cut is

$$\omega^M = \{0^M, 1^M, 2^M, \dots\}.$$

We identify standard numerals with ordinary natural numbers when no confusion can result. The interval $[0, a)$ always uses the order of M . If a is standard it is externally finite; otherwise it is externally countably infinite.

A *cut* in this report is an external subset $I \subseteq M$ containing zero, closed downward and under successor. Thus $\omega^M \subseteq I$, and we allow $I = M$. No closure under addition or multiplication is assumed. Every cut has no greatest element. Any two cuts are comparable by inclusion. An external set need not be definable in the original arithmetic structure.

The model has a unique interpretation of the provably total function $E(t) = 2^t$. We may name it in a definitional expansion. The bit relation is arithmetically definable; with E named one convenient definition is

$$\text{Bit}(c, t) \iff \exists q \leq c \exists r < E(t) (c = qE(t+1) + E(t) + r). \quad (2.1)$$

All arithmetic in this formula takes place in M . Naming E is only a convenience for syntax; later complexity claims distinguish the named exponential language from the bare arithmetic language.

Definition 2.1 (Coding cut). For $P \subseteq M$, define

$$\theta_P(a) : \exists c < E(a) \forall t < a (P(t) \leftrightarrow \text{Bit}(c, t)), \quad (2.2)$$

$$\text{Cut}(P) = \{a \in M : (M, P) \models \theta_P(a)\}. \quad (2.3)$$

Let $\mathcal{D}_a$ be the family of subsets of $[0, a)$ that have such a code.

The formula is fixed independently of the model, the biases, and the desired cut. It has no parameters besides its free endpoint a and the predicate symbol. A law used to sample the predicate is external data, not an extra symbol in this formula.

Lemma 2.2 (Elementary coding operations). *For every $a \in M$, the family $\mathcal{D}_a$ contains the empty set and every externally finite subset of $[0, a)$. It is closed under union, intersection, complement within $[0, a)$, and symmetric difference. Restrictions of coded segments are coded. A coded segment below a can be extended by either bit at a to a coded segment below $a + 1$.*

Proof. For externally finitely many distinct positions $t_1, \dots, t_k < a$, the internal integer $\sum_{j=1}^k E(t_j)$ is a code. The positions themselves may be nonstandard; only the number k of operations in this construction is standard finite. The usual bit-by-bit Boolean operations are provably total primitive recursive functions in PA. Equivalently, PA proves bounded coding for each of the arithmetic formulas obtained by combining the bit predicates of finitely many given codes. Restriction to a shorter interval is reduction modulo its power of two. Finally, if $c < E(a)$ is a code below a , the extensions are c and $c + E(a)$, both below $E(a + 1)$. These arguments involve only arithmetic induction in M ; they use no induction involving the external predicate P . $\square$

Proposition 2.3 (Deterministic structure). *For every predicate P , $\text{Cut}(P)$ is a cut. Moreover:*

- (i) *changing finitely many bits below a does not change whether $a \in \text{Cut}(P)$;*
- (ii) $\text{Cut}(P) = \text{Cut}(M \setminus P)$;
- (iii) $\text{Cut}(P) \cap \text{Cut}(Q) \subseteq \text{Cut}(P \triangle Q)$;
- (iv) *if $\text{Cut}(P) \neq \text{Cut}(Q)$, then $\text{Cut}(P \triangle Q) = \text{Cut}(P) \cap \text{Cut}(Q)$.*

Proof. Every standard initial segment is a finite pattern, so it is coded. Downward closure and successor closure follow from Lemma 2.2. The same lemma proves (i)–(iii). For (iv), suppose $I = \text{Cut}(P) \subsetneq J = \text{Cut}(Q)$. At any $a \in J \setminus I$, the segment of Q is coded and that of P is not. Coding their symmetric difference would therefore code P , a contradiction. For an arbitrary $b \notin I$, choose $a \in J \setminus I$. If $b \in J$, apply the preceding argument directly to b . Otherwise $a < b$, so downward closure excludes b from the symmetric-difference cut as well. This proves equality everywhere. $\square$

Remark 2.4 (A filtration by cuts). Under symmetric difference, predicates form a vector space over $\mathbb{F}_2$. The predicates coded below a fixed endpoint form a subspace. Proposition 2.3 gives the familiar strong-triangle behavior of a filtration: two different coding cuts cannot cancel to give a larger one. If $\text{Cut}(R) = M$, then $\text{Cut}(P \triangle R) = \text{Cut}(P)$ for every P . Thus finite changes are just one class of harmless perturbations; every predicate coded on every bounded interval is harmless under symmetric difference.

For a model of PA, being coded on every bounded interval is equivalent to being a *class*: each bounded restriction is definable with parameters. One direction uses its binary code; the other uses arithmetic bounded coding of a definable set. This matches the class terminology in the literature on generic expansions [2].

3 The exact classification for independent noise

For a family $w = (w_t)_{t \in M}$ of finite nonnegative real numbers, write

$$W_w(a) = \sum_{t < a} w_t = \sup \left\{ \sum_{t \in F} w_t : F \subseteq [0, a) \text{ is externally finite} \right\}, \quad S(w) = \{a : W_w(a) < \infty\}. \quad (3.1)$$

This is an ordinary real sum in the metatheory. It is not a nonstandard sum calculated by M . Adding the one summand at a preserves finiteness, so $S(w)$ is always a cut.

Give 2^M its external product topology and its Borel sigma-algebra. As M is countable, this is a Cantor space after choosing an external enumeration. For biases $p_t \in [0, 1]$, let $\mu_p = \bigotimes_{t \in M} \text{Bern}(p_t)$.

Put

$$\varepsilon_t = \min(p_t, 1 - p_t), \quad b_t = \mathbf{1}_{\{p_t > 1/2\}}. \quad (3.2)$$

The tie convention at $p_t = 1/2$ is immaterial to the theorem.

Theorem 3.1 (Arbitrary independent biases). *For every countable $M \models \text{PA}$ and every family of independent Bernoulli biases, with probability one simultaneously at all endpoints,*

$$\boxed{\text{Cut}(P) = \text{Cut}(b) \cap S(\varepsilon).} \quad (3.3)$$

In particular, when $0 \leq p_t \leq 1/2$ for all t ,

$$\boxed{\text{Cut}(P) = S(p) \text{ almost surely.}} \quad (3.4)$$

Proof. Fix an endpoint a . Suppose first that $\sum_{t < a} \varepsilon_t < \infty$. The probability of a mismatch with b at t is exactly ε_t . Enumerate $[0, a)$ externally. The probability of any mismatch after the first L positions is at most the sum of their mismatch probabilities, which tends to zero as $L \rightarrow \infty$. Consequently only finitely many mismatches occur almost surely. By Proposition 2.3(i), P is coded below a if and only if b is coded there. This half needs no independence.

Suppose instead that $\sum_{t < a} \varepsilon_t = \infty$. For any fixed pattern d on $[0, a)$ and any externally finite $F \subseteq [0, a)$, independence gives

$$\mu_p\{P|_F = d|_F\} \leq \prod_{t \in F} (1 - \varepsilon_t) \leq \exp\left(-\sum_{t \in F} \varepsilon_t\right). \quad (3.5)$$

As the finite sets exhaust the interval, the right side tends to zero. Thus every specified pattern has probability zero. Since M is countable, the internal code family $\mathcal{D}_a$ is countable, so the probability of matching any internal code is zero.

Both alternatives give the asserted equivalence at the fixed endpoint. Intersecting the corresponding probability-one events over the countably many $a \in M$ proves the simultaneous assertion. If all $p_t \leq 1/2$, the chosen modal predicate is empty, hence has coding cut M . $\square$

Corollary 3.2 (The fair-coin case). *Under fair independent bits, $\text{Cut}(P) = \omega^M$ almost surely. More generally, the same holds whenever $\sum_{t < a} \min(p_t, 1 - p_t) = \infty$ for every nonstandard a ; in particular it holds if all biases lie in $[\delta, 1 - \delta]$ for one $\delta > 0$.*

Proof. Every nonstandard interval is externally infinite. Fair bits contribute $1/2$ at every position, and uniformly nondegenerate bits contribute at least δ . Standard intervals have finite sums. Apply Theorem 3.1. $\square$

This corollary recovers the PA conclusion announced by Glazer; the classification of all nonuniform product laws is the added issue here. Countability is used twice: to have countably many codes at each endpoint, and to combine the conclusions for all endpoints. We make no extension of this proof to arbitrary cardinalities. The uncountable measurability issues raised in Glazer's talk remain separate.

3.1 Three elementary examples

Example 3.3 (Globally finite noise). Enumerate $M = \{m_0, m_1, \dots\}$ externally and set $p_{m_j} = 2^{-j-3}$. The total mass is finite, so almost surely P is an externally finite set and $\text{Cut}(P) = M$. Every finite bit pattern still has positive probability. The law is atomic despite having full topological support.

Example 3.4 (Noise located on the standard cut). Set $p_n = c/(n+2)^\alpha$ at standard positions, with $0 < c \leq 1/4$, and choose positive, globally summable biases off the standard cut. If $\alpha \leq 1$, every nonstandard interval sees divergent mass and $\text{Cut}(P) = \omega^M$. If $\alpha > 1$, the total mass is finite and $\text{Cut}(P) = M$. These are external probability prescriptions, not formulas defining the standard cut inside the original structure.

Example 3.5 (The mode matters). Given any fixed external predicate B , flip its bits independently with globally summable error probabilities. Theorem 3.1 gives $\text{Cut}(P) = \text{Cut}(B)$ almost surely. Small total noise does not make a noncodable modal pattern codable.

4 Realizing any cut without changing prescribed finite data

Lemma 4.1 (Cofinal markers). *For each cut $I \subseteq M$ and finite set $F \subseteq M$, there is an externally strictly increasing sequence $u_0 < u_1 < \dots$ in $I \setminus F$ cofinal in I . Its set U satisfies*

$$|U \cap [0, a)| < \infty \quad (a \in I), \quad |U \cap [0, a)| = \infty \quad (a \notin I).$$

Proof. Enumerate I externally. At step n , choose a point of I above the first n enumerated points and the preceding marker, avoiding F . A cut is closed under successor, so finitely many exclusions never prevent this. For $a \in I$, some marker is at least a and all later ones are larger. For $a \notin I$, downward closure forces every point of I to lie below a . This also proves the claim when $I = M$; the second alternative is then vacuous. $\square$

Theorem 4.2 (Arbitrary cut, finite marginals, full support). *Fix a cut I , a finite set $F \subseteq M$, and prescribed numbers $q_t \in (0, 1)$ for $t \in F$. There exists an independent product law with the following properties:*

- (i) $p_t = q_t$ on F and $0 < p_t < 1$ everywhere;
- (ii) the law has full topological support and no atoms;
- (iii) $\text{Cut}(P) = I$ almost surely.

Outside F , the probabilities may all be taken at most $1/4$.

Proof. Take cofinal markers $U \subseteq I \setminus F$ as in Lemma 4.1. Set $p_t = 1/4$ on U , keep the prescribed values on F , and assign the remaining points strictly positive probabilities less than $1/4$ with finite total sum. For example, an external enumeration assigns them 2^{-j-4} .

The minority-mass sum below a is finite exactly when $a \in I$: finitely many prescribed coordinates and summable residual noise do not affect divergence, and the markers have precisely the required intersections. The modal predicate is supported on a subset of F , hence is internally coded everywhere. Theorem 3.1 now gives the required cut.

Every nonempty basic cylinder fixes only finitely many bits and therefore has positive probability, proving full support. For any fixed full predicate, its probability is bounded by $(3/4)^k$ on any k

markers; letting $k \rightarrow \infty$ proves nonatomicity. This argument includes $I = M$: global randomness can be infinite although every bounded interval has finite noise mass. $\square$

Corollary 4.3 (Weak limits do not preserve the coding cut). *For any cut I and any enumeration $m_0, m_1, \dots$ of M , there are full-support nonatomic product laws μ_n for which*

$$\mu_n\{\text{Cut}(P) = I\} = 1,$$

whose first n enumerated coordinates are exactly independent fair bits, and which converge weakly to fair product measure μ . If $I \neq \omega^M$, then $\mu\{\text{Cut}(P) = I\} = 0$ and $\text{TV}(\mu_n, \mu) = 1$ for every n , where TV is the supremum of the difference on measurable events.

Proof. Apply Theorem 4.2 with $F_n = \{m_0, \dots, m_{n-1}\}$ and $q_t = 1/2$. Every fixed finite cylinder has its fair probability for all sufficiently large n . On the compact space 2^M , continuous functions are uniformly approximable by functions depending on finitely many coordinates: use uniform continuity in any compatible product metric. Thus cylinder convergence proves weak convergence. The exact-cut event is Borel by Proposition 7.1 below. It has probabilities one and zero under the two laws when $I \neq \omega^M$, proving the total-variation assertion. $\square$

The quantifiers matter. For each chosen finite set a law can be selected that agrees with fair coins there. A single law agreeing with fair coins on *every* finite set is the fair product law itself. The theorem does not evade that uniqueness fact. It says that no preassigned finite collection of marginal data determines the infinite definability outcome.

5 Independent predicates and complete Boolean coding spectra

The product classification has a stronger consequence than the realization of a single cut. It permits a complete description of the cuts obtained by taking all finite conjunctions of a countable family of independent predicates whose biases are uniformly bounded away from one. In this regime the only restriction is inclusion monotonicity. Once these conjunction cuts are specified, the coding cut of every finitary Boolean observable is determined. All statements in this section concern the fixed countable model $M \models \text{PA}$. The index set $\mathbb{N}$ is external and standard.

For a nonnegative external weight $w : M \rightarrow [0, \infty)$, recall the notation

$$S(w) = \left\{ a \in M : \sum_{x < a} w(x) < \infty \right\}.$$

Every sum here is an external countable sum. Throughout the section, a *cut* means a nonempty initial segment of M closed under successor; the whole of M is allowed. In particular, every such cut has no greatest element. A family of Bernoulli predicates is called independent when all its bits, indexed by both the predicate and the element of M , are mutually independent.

5.1 A useful one-sided form of the product classification

Lemma 5.1 (Bias bounded away from one). *Let $X \subseteq M$ have independent bits with $\mathbb{P}(x \in X) = q(x)$. If a constant $\eta > 0$ satisfies $q(x) \leq 1 - \eta$ for every $x \in M$, then*

$$\text{Cut}(X) = S(q) \quad \text{almost surely.}$$

Proof. We have the pointwise comparisons

$$\eta q(x) \leq \min\{q(x), 1 - q(x)\} \leq q(x).$$

For the first inequality, use $\min\{q, 1 - q\} = q$ when $q \leq 1/2$; when $q > 1/2$, use $1 - q \geq \eta \geq \eta q$. If $\sum_{x < a} q(x) = \infty$, Theorem 3.1 therefore excludes a from the coding cut almost surely. If the sum is finite, the number of ones of X below a is finite almost surely: its expectation is that sum, or equivalently the first Borel–Cantelli lemma applies. A finite external set of positions below a has an internal binary code. Finally, there are only countably many $a \in M$, so both conclusions hold simultaneously. $\square$

Thus probabilities may pass above $1/2$ in this lemma. The uniform separation from 1 prevents a second, competing baseline from introducing a new coding obstruction. In the constructions below all probabilities can be kept below an arbitrarily small fixed positive bound.

5.2 Universality for all finite conjunctions

Write

$$\mathcal{F} = \{S \subseteq \mathbb{N} : 0 < |S| < \infty\}, \quad P_S = \bigcap_{i \in S} P_i.$$

An assigned family $(C_S)_{S \in \mathcal{F}}$ of cuts is *inclusion monotone* if

$$S \subseteq T \implies C_S \subseteq C_T. \quad (5.1)$$

The direction is significant: intersecting more independent predicates can make their bounded restrictions easier to code.

Theorem 5.2 (Countable conjunction universality). *Let $(C_S)_{S \in \mathcal{F}}$ be any family of cuts of M , and fix $0 < \delta < 1/2$. The following are equivalent.*

1. *The family satisfies (5.1).*
2. *There is an independent Bernoulli array $(P_i)_{i \in \mathbb{N}}$, with probabilities $0 < p_i(x) < \delta$ for every i, x , such that*

$$\mathbb{P}(\text{Cut}(P_S) = C_S \text{ for every } S \in \mathcal{F}) = 1. \quad (5.2)$$

The realizing measure can have full support and no atoms. Moreover, it can be chosen so that almost surely every P_S is infinite. Whenever $C_S = M$, that P_S is cofinal in M and has only finitely many elements below each $a \in M$.

Proof. Necessity. The bits of P_S are independent, with probabilities

$$p_S(x) = \prod_{i \in S} p_i(x).$$

Lemma 5.1 gives $\text{Cut}(P_S) = S(p_S)$ almost surely. If $S \subseteq T$, then $p_T(x) \leq p_S(x)$, whence $S(p_S) \subseteq S(p_T)$. Countability of $\mathcal{F}$ makes these statements simultaneous and proves the required condition.

The diagonal marker construction. Assume (5.1). Enumerate all constraints

$$\{(S, a) : S \in \mathcal{F}, a \in C_S\} = \{(S_j, a_j) : j \in \mathbb{N}\}.$$

Choose a sequence $(U_n)_{n \in \mathbb{N}}$ in $\mathcal{F}$ in which each $U \in \mathcal{F}$ occurs infinitely often. At stage n , select a fresh point $x_n \in C_{U_n}$ above every a_j such that $j \leq n$ and $S_j \subseteq U_n$. This is possible: each such a_j belongs to C_{U_n} by monotonicity, and there are only finitely many such bounds. To ensure freshness, also put x_n above each previously selected point that belongs to C_{U_n} . These are again finitely many points of that cut. The successor of their maximum, including 0 among the bounds if necessary, belongs to C_{U_n} and is an admissible choice. Previously selected points outside C_{U_n} cannot equal this new point.

For $S \in \mathcal{F}$, define its marker set

$$A_S = \{x_n : S \subseteq U_n\}.$$

We claim that

$$A_S \cap [0, a) \text{ is finite} \iff a \in C_S. \quad (5.3)$$

If $a \in C_S$, find an index j of the constraint (S, a) . For every $n \geq j$ with $S \subseteq U_n$, the construction imposes $x_n > a$. Only finitely many earlier markers can lie below a . Conversely, if $a \notin C_S$, every marker with label $U_n = S$ belongs to C_S and hence lies below a . There are infinitely many such stages, and all their markers are distinct. This proves (5.3).

The diagonal requirement is needed for a countable family. Choosing individually cofinal marker sequences for the cuts would control each sequence separately, but an infinite union of these sequences could still have infinitely many points below a member of C_S .

Probabilities and products. Choose a strictly positive, globally summable function $\rho : M \rightarrow (0, \delta/4]$, and put $c = \delta/2$. For example, for an external enumeration $M = \{z_0, z_1, \dots\}$ one may use $\rho(z_n) = \delta 2^{-n-2}$. Define

$$p_i(x) = \rho(x) + c \mathbf{1}_{\{x=x_n \text{ for some } n \text{ with } i \in U_n\}}. \quad (5.4)$$

Then $0 < p_i(x) \leq 3\delta/4 < \delta$. Since the marker points are distinct, at most one marker label contributes at any given x .

For $S \in \mathcal{F}$ and $s = |S|$, expansion of the product gives

$$p_S(x) = c^s \mathbf{1}_{A_S}(x) + E_S(x), \quad 0 \leq E_S(x) \leq s\rho(x). \quad (5.5)$$

Here is a direct verification of the error bound. At a marker whose label contains S , the product is $(c + \rho(x))^s$, and telescoping its difference from c^s bounds that difference by $s\rho(x)$, because all factors are at most 1. At any other point at least one of the factors equals $\rho(x)$, so the product itself is at most $\rho(x)$.

The error is globally summable. Consequently

$$S(p_S) = S(\mathbf{1}_{A_S}) = C_S,$$

where the last equality is (5.3). Take the countable product measure under which all bits have the prescribed probabilities. Lemma 5.1, followed by the intersection of countably many probability-one events, proves (5.2).

Full support, absence of atoms, and infinite outputs. Every finite cylinder in $\{0, 1\}^{\mathbb{N} \times M}$ has positive probability because each coordinate probability lies strictly between 0 and 1. For any fixed i , there are infinitely many markers labeled $\{i\}$. At each of them,

$$\min\{p_i(x), 1 - p_i(x)\} = p_i(x) \geq c.$$

The probability of matching any prescribed bit sequence at the first k such markers is at most $(1 - c)^k$. Thus each marginal measure, and hence the entire array measure, has no atoms.

For a fixed S , the infinitely many markers labeled exactly S belong to P_S independently, each with probability at least $c^{|S|}$. The second Borel–Cantelli lemma gives infinitely many successes almost surely. This holds for every S simultaneously. If $C_S = M$, then $\sum_{x < a} p_S(x) < \infty$ for every $a \in M$, so $P_S \cap [0, a)$ is finite almost surely, simultaneously for all a . An infinite subset with that property cannot be bounded in M , and is therefore cofinal. $\square$

Remark 5.3 (Scope of the necessity statement). Condition (5.1) is necessary for this independent representation with uniformly small probabilities. It is not a general law for intersections of arbitrary deterministic predicates. If $\text{Cut}(A) \neq M$ and $B = M$ is the constantly true predicate, then $A \cap B = A$ while $\text{Cut}(B) = M$. Also, A and its complement have the same coding cut, although their union and symmetric difference are both M . Independence and the bounds on the probabilities are substantive hypotheses of the spectral laws.

Remark 5.4 (A finite family). The same proof applies to a finite index set, because its collection of constraints is still countable. Thus any prescribed finite table of conjunction cuts is realizable precisely when it satisfies (5.1). The countable theorem supplies a common probability-one event for all finite conjunctions. It does not assert a corresponding law for countably infinite conjunctions or disjunctions.

5.3 Every finitary Boolean observable

Let $F \subseteq \mathbb{N}$ be finite and nonempty, and let $f : \{0, 1\}^F \rightarrow \{0, 1\}$ be a Boolean function. For $S \subseteq F$, write $\mathbf{1}_S$ for the vector with ones exactly on S , and set $b_f = f(\mathbf{0})$. Define

$$\mathcal{D}(f) = \min_{\subseteq} \{S \subseteq F : f(\mathbf{1}_S) \neq b_f\}. \quad (5.6)$$

These are the inclusion-minimal sets of successes that change the output from its all-zero value. They are nonempty subsets of F . $\mathcal{D}(f)$ is empty precisely when f is constant.

The observable predicate is

$$f(P) = \{x \in M : f((\mathbf{1}_{P_i}(x))_{i \in F}) = 1\}.$$

Theorem 5.5 (Boolean classification). *Suppose the array is independent and $0 \leq p_i(x) \leq \delta < 1$ for every $i \in F$ and $x \in M$. Then*

$$\text{Cut}(f(P)) = \bigcap_{S \in \mathcal{D}(f)} S \left(\prod_{i \in S} p_i \right) = \bigcap_{S \in \mathcal{D}(f)} \text{Cut}(P_S) \quad \text{almost surely.} \quad (5.7)$$

An empty intersection is interpreted as M . For countably many independent predicates obeying a common bound $\delta < 1$, these identities hold simultaneously for all finitary Boolean functions.

Proof. Constants have coding cut M , so suppose f is nonconstant. Let $r = |F|$, $\eta = (1 - \delta)^r$, and define the deviation probability

$$e_f(x) = \mathbb{P}(f((\mathbf{1}_{P_i}(x))_{i \in F}) \neq b_f).$$

The event that all r input bits are zero has probability at least η , and never contributes to a deviation. Hence

$$0 \leq e_f(x) \leq 1 - \eta. \quad (5.8)$$

We also have the two-sided estimate

$$\eta \sum_{S \in \mathcal{D}(f)} \prod_{i \in S} p_i(x) \leq e_f(x) \leq \sum_{S \in \mathcal{D}(f)} \prod_{i \in S} p_i(x). \quad (5.9)$$

For the upper bound, any success set on which f deviates contains an inclusion-minimal deviating subset. A union bound over the events that all coordinates of such an S equal one gives the assertion. This argument does not assume that f is monotone.

For the lower bound, consider the disjoint events that the exact success set in F is a specified $S \in \mathcal{D}(f)$. Each is a deviating event with probability

$$\prod_{i \in S} p_i(x) \prod_{j \in F \setminus S} (1 - p_j(x)) \geq \eta \prod_{i \in S} p_i(x).$$

Summing these disjoint contributions gives the lower inequality in (5.9).

Let D_f be the predicate of deviations from b_f . Its bits are independent. If $b_f = 0$, then $D_f = f(P)$; if $b_f = 1$, they are complements. Complementation preserves coding cuts, since the complement of an internally coded string of length a is internally coded as well. Lemma 5.1 and (5.8) therefore give $\text{Cut}(f(P)) = S(e_f)$ almost surely. By (5.9), a sum of e_f over an initial segment is finite exactly when every one of the finitely many corresponding product sums is finite. This proves the first equality in (5.7); the same lemma applied to each P_S proves the second. There are only countably many Boolean functions on finite subsets of $\mathbb{N}$, so all the equalities can be imposed simultaneously. $\square$

Together, Theorems 5.2 and 5.5 give a complete realization statement: specify any inclusion-monotone table (C_S) , and the entire finitary Boolean spectrum is

$$f \mapsto \bigcap_{S \in \mathcal{D}(f)} C_S. \quad (5.10)$$

Conversely, every independent array whose probabilities are uniformly bounded above by a constant less than 1 has this form for its own table of conjunction cuts. The construction proves realizability even under the stronger requirement that all probabilities lie below an arbitrarily prescribed $\delta < 1/2$.

Corollary 5.6 (Nonconstant observables remain infinite). *In the countable realization of Theorem 5.2, almost surely every nonconstant finitary Boolean observable takes each of the values 0 and 1 at infinitely many points of M .*

Proof. Fix f on F and choose $S \in \mathcal{D}(f)$. There are infinitely many markers labeled exactly S . At each, the probability of the exact success pattern S within F is at least $c^{|S|}(1 - \delta)^{|F \setminus S|} > 0$. Independent repetitions give infinitely many deviations from b_f almost surely. Choose an index $j \notin F$. At the infinitely many markers labeled $\{j\}$, the probability that all inputs in F are zero is at least $(1 - \delta)^{|F|} > 0$. These give infinitely many occurrences of b_f . There are countably many f , so the assertion is simultaneous. $\square$

In particular, a maximal coding cut in the universality theorem can be carried by a nonconstant infinite predicate. It need not result from an observable that has only finitely many exceptional values.

5.4 Examples and sharp operation laws

Union and symmetric difference. For two predicates P, Q , the minimal sensitive supports of both OR and XOR are the two singletons. Consequently

$$\text{Cut}(P \cup Q) = \text{Cut}(P \triangle Q) = \text{Cut}(P) \cap \text{Cut}(Q) \quad \text{almost surely} \quad (5.11)$$

under the hypotheses of Theorem 5.5. For example, when $p, q \leq 1/4$, the coordinate probabilities are $p + q - pq$ and $p + q - 2pq$, respectively; each is comparable to $p + q$. For parity or union of any fixed finite number of inputs, the same singleton-support argument gives the intersection of all input cuts.

Intersection and its complete three-cut range. For AND the only minimal sensitive support is the full set of inputs. If I, J, K are cuts, there exist independent predicates with uniformly arbitrarily small positive biases satisfying

$$\text{Cut}(P) = I, \quad \text{Cut}(Q) = J, \quad \text{Cut}(P \cap Q) = K \quad \text{almost surely} \quad (5.12)$$

if and only if $I \cup J \subseteq K$. This is the two-input instance of Theorem 5.2. Both $K = I \cup J$ and $K = M$ are allowed, as is every intermediate cut. Because cuts of a linear order are nested, $I \cup J$ is their larger member and $I \cap J$ their smaller member.

A direct construction makes this case particularly transparent. Choose pairwise disjoint increasing cofinal marker sequences $A \subseteq I$, $B \subseteq J$, and $C \subseteq K$. They can be placed in the three residue classes modulo 3: every successor cut contains arbitrarily large points in each fixed residue class relative to its own elements. With $c > 0$ small and a positive globally summable ρ , put

$$p = \rho + c\mathbf{1}_{A \cup C}, \quad q = \rho + c\mathbf{1}_{B \cup C}.$$

The summability cuts of these weights are $I \cap K = I$ and $J \cap K = J$, while

$$pq = c^2\mathbf{1}_C + c\rho(\mathbf{1}_A + \mathbf{1}_B + 2\mathbf{1}_C) + \rho^2.$$

The last two terms are globally summable, giving product cut K . This construction also works when one or more of the target cuts equal M .

Thresholds and majority. Let Thr_t output one when at least t of r bits equal one. Its minimal sensitive supports are exactly the t -element subsets, so

$$\text{Cut}(\text{Thr}_t(P_1, \dots, P_r)) = \bigcap_{\substack{S \subseteq \{1, \dots, r\} \\ |S|=t}} C_S \quad \text{almost surely.} \quad (5.13)$$

Given any increasing chain of cuts $K_1 \subseteq K_2 \subseteq \dots \subseteq K_r$, assign $C_S = K_{|S|}$ in the finite-family theorem. Then

$$\text{Cut}(\text{Thr}_t(P_1, \dots, P_r)) = K_t \quad (1 \leq t \leq r)$$

simultaneously. Thus every possible increasing hierarchy of threshold cuts is realized. For odd r , majority is the case $t = (r + 1)/2$.

A nonmonotone example. For $f(u, v) = u \wedge \neg v$, the only minimal sensitive support is $\{1\}$, even though the support $\{1, 2\}$ returns the function to its baseline. Hence

$$\text{Cut}(P \setminus Q) = \text{Cut}(P) \quad \text{almost surely}$$

under the uniform bound on the input probabilities. At the level of weights this is the comparison $(1 - \delta)p \leq p(1 - q) \leq p$.

5.5 A series threshold for repeated independent thinning

Proposition 5.7 (Power-law marker threshold). *Let I be a proper cut of M and let $\alpha > 0$. There are independent, identically distributed predicates $(P_i)_{i \in \mathbb{N}}$ with strictly positive uniformly small probabilities such that, simultaneously for every nonempty finite $S \subseteq \mathbb{N}$,*

$$\text{Cut}(P_S) = \begin{cases} I, & |S|^\alpha \leq 1, \\ M, & |S|^\alpha > 1 \end{cases} \quad \text{almost surely.}$$

In the second case P_S is externally finite almost surely.

Proof. Choose an increasing cofinal sequence $(x_n)_{n \in \mathbb{N}}$ in I , a small $c > 0$, and a strictly positive globally summable noise ρ small enough that the following probabilities are below $1/4$:

$$p_i(x) = p(x) = \begin{cases} \rho(x_n) + c(n+1)^{-\alpha}, & x = x_n, \\ \rho(x), & x \notin \{x_n : n \in \mathbb{N}\}. \end{cases}$$

For $r = |S|$, the weight p^r differs by a globally summable nonnegative error, bounded pointwise by $r\rho$, from the marker weight $c^r(n+1)^{-r\alpha}$. Below any $a \in I$ there are only finitely many markers. Below any $a \notin I$ there are all of them. The ordinary positive-series test therefore gives summability cut I when $r\alpha \leq 1$, including the harmonic boundary, and M when $r\alpha > 1$. In the latter case the sum of the success probabilities over the whole of M is finite, giving the asserted finiteness of the random set. Lemma 5.1 and countability finish the proof. $\square$

For example, taking $\alpha = 1/k$ leaves the coding cut equal to I through conjunction order k , and makes it maximal at order $k+1$. Theorem 5.2 allows a more general prescribed hierarchy, and realizes maximal cuts by infinite cofinal outputs as well.

5.6 Universality with vanishing probabilities

Proposition 5.8 (Vanishing-bias refinement). *Theorem 5.2 remains true with the additional condition that, for each $\varepsilon > 0$, the set*

$$\{x \in M : \sup_{i \in \mathbb{N}} p_i(x) \geq \varepsilon\}$$

is finite. Full support, absence of atoms, and the infinite-output conclusions can all be retained.

Proof. Enumerate the marker labels as $V_0, V_1, \dots$. Keep the diagonal marker placement unchanged, but at the k th occurrence of label V_m replace the constant c in (5.4) by

$$d_{m,k} = \frac{c 2^{-m}}{1 + \log(k+1)}, \quad m, k \geq 0,$$

so that $0 < d_{m,k} \leq c$. Only finitely many pairs (m, k) have $d_{m,k} \geq \varepsilon$ for any fixed $\varepsilon > 0$. A globally summable ρ has the same finiteness property, proving the claimed vanishing condition.

For each fixed label V_m and each positive integer s ,

$$\sum_{k \geq 0} d_{m,k}^s = \infty.$$

Indeed, $(1 + \log(k + 1))^{-s} \geq 1/k$ for all sufficiently large k . For a fixed S , the leading term of the product weight is now $d_{m,k}^{|S|}$ at markers whose label V_m contains S , and zero elsewhere. The error remains bounded by $|S|\rho$. On C_S its marker support is locally finite by (5.3); outside C_S , the markers with label exactly S already have divergent total leading weight. Thus the summability cut is still C_S .

For each input coordinate, the sum of the smaller atom probabilities along its singleton-labeled markers diverges, so no atom can occur. For each finite conjunction, the corresponding success probabilities along its own label also have divergent sum; independence gives infinitely many successes. The same argument, with the fixed lower bound on complementary factors, proves the infinite-deviation assertion for a nonconstant Boolean observable. The proofs of full support and of cofinality when the cut is M are unchanged. $\square$

The condition in Proposition 5.8 is invariant under the choice of an external enumeration of M . It describes vanishing outside finite external sets. It does not assert that the probabilities vanish as x increases in the internal order of M .

6 An exact induction phase diagram

The coding cut also measures a precise fragment of induction in the expanded arithmetic. Throughout this section, M is a fixed countable *nonstandard* model of PA, unless a statement explicitly says otherwise. The function $E(x) = 2^x$ is named in a definitional expansion of the arithmetic language. Induction for formulas that contain the new predicate P is never assumed in a coding argument.

6.1 Keeping track of the language

In the language with E , the formula

$$\vartheta(x, P) \equiv \exists c < E(x) \forall a < x (P(a) \leftrightarrow \text{Bit}(c, a)) \quad (6.1)$$

is bounded. For example, its bit relation can be expanded as

$$\text{Bit}(c, a) \equiv \exists q \leq c \exists r < E(a) (c = qE(a + 1) + E(a) + r). \quad (6.2)$$

Thus the precise complexity assertion is $\vartheta \in \Delta_0(P, E)$. Naming a definable function is harmless for first-order definability, but does not automatically preserve a bounded-quantifier classification in the original language.

For a bare-language formulation, put

$$\text{Rem}(u, m, r) \equiv r < m \wedge \exists q \leq u (u = qm + r), \quad (6.3)$$

$$\text{Beta}(u, v, i, r) \equiv \text{Rem}(u, 1 + (i + 1)v, r). \quad (6.4)$$

These are bounded formulas in the language of addition and multiplication. Define

$$\chi(x, P) \equiv \exists u \exists v \left[v > 0 \wedge \forall i < x \left((P(i) \wedge \text{Beta}(u, v, i, 1)) \vee (\neg P(i) \wedge \text{Beta}(u, v, i, 0)) \right) \right]. \quad (6.5)$$

The matrix following $\exists u \exists v$ is bounded, so this is a $\Sigma_1(P)$ formula in the original arithmetic language.

Lemma 6.1 (Two equivalent coding formulas). *For every predicate $P \subseteq M$ and every $x \in M$,*

$$(M, P) \models \chi(x, P) \iff (M, P) \models \vartheta(x, P).$$

The equivalence uses arithmetic coding theorems of PA, without induction involving P .

Proof. The standard finite-sequence coding theorem of PA converts a binary code into a Beta code for the same internally finite sequence. Here is the arithmetic construction underlying this use of the theorem. For a sequence of length x whose entries are 0 or 1, choose a positive v divisible by $x!$ and larger than 1. The moduli

$$1 + v, 1 + 2v, \dots, 1 + xv$$

are pairwise coprime. Indeed, a common divisor of $1 + (i + 1)v$ and $1 + (j + 1)v$ is coprime to v and divides $j - i$; as $j - i$ divides $x!$, it also divides v , forcing that common divisor to be 1. The internally finite Chinese remainder theorem supplies u with the prescribed binary residues. For the sequence given by the digits of a fixed arithmetic code c , this entire construction, including the finite Chinese remainder theorem, is proved in PA by ordinary arithmetic induction. No occurrence of P is needed in that arithmetic proof. A witness to ϑ can consequently be converted into witnesses to χ .

Conversely, PA proves that a Beta code of length x with all entries in $\{0, 1\}$ can be converted to the binary code

$$c = \sum_{i < x} r_i E(i), \quad \text{Beta}(u, v, i, r_i).$$

This displayed sum abbreviates the primitive recursive arithmetic construction of its partial sums; it is not an external sum over a nonstandard interval. Ordinary arithmetic induction proves that the construction is total, that $c < E(x)$, and that its digits are the r_i . When u, v witness χ , their residues are already required to be 0 or 1 and to agree with P on $[0, x)$. The resulting c therefore witnesses ϑ . In both directions the arithmetic construction takes numerical codes as its inputs; P is used only to identify the pattern those codes represent. $\square$

Corollary 6.2. *If $\text{Cut}(P)$ is proper, then (M, P) fails $\text{I}\Sigma_1(P)$ in the original arithmetic language.*

Proof. The formula χ defines $\text{Cut}(P)$ by Lemma 6.1. This set contains 0 and is closed under successor, but is not all of M . Consequently the induction instance for the $\Sigma_1(P)$ formula χ is false. $\square$

6.2 Bounded induction is equivalent to local coding

Write $\text{I}\Delta_0(P, E)$ for induction for every bounded formula in the language with P and the named function E , allowing parameters from M .

Theorem 6.3 (Local coding and bounded induction). *For an arbitrary predicate $P \subseteq M$,*

$$\text{Cut}(P) = M \iff (M, P) \models \text{I}\Delta_0(P, E). \quad (6.6)$$

This equivalence also holds when M is standard, and does not require countability of M .

Proof. Suppose first that bounded induction holds. The bounded formula ϑ satisfies $\vartheta(0, P)$. If $c < E(x)$ codes the predicate below x , then either c or $c + E(x)$ codes it below $x + 1$, according as $P(x)$ is false or true. The arithmetic facts about this extension of a binary code are theorems of PA. Therefore

$$(M, P) \models \vartheta(0, P) \wedge \forall x (\vartheta(x, P) \rightarrow \vartheta(x + 1, P)).$$

Induction for ϑ gives $\forall x \vartheta(x, P)$, which says $\text{Cut}(P) = M$.

For the converse, assume $\text{Cut}(P) = M$. Let $\delta(z, P, \bar{a})$ be an arbitrary bounded formula, with a fixed tuple of parameters $\bar{a}$ from M . Suppose its induction premises hold in (M, P) , and suppose toward a contradiction that $\neg\delta(b, P, \bar{a})$ for some $b \in M$.

All occurrences of P needed to evaluate this particular formula for $z \leq b + 1$ have arguments below one element $T \in M$. This assertion is a bound on a fixed finite formula, not a use of induction in the expanded model. To see it, majorize the free variables by $b + 1 + \sum_j a_j$, increasing the bound by 1 if needed. The arithmetic function symbols $0, S, +, \cdot, E$ are monotone on nonnegative arguments. At each bounded quantifier, substitute the already established bounds into its bounding term to obtain a bound for the new variable. There are only finitely many nested quantifiers and terms. Repeating this syntactic construction and then majorizing the finitely many argument terms of predicate occurrences gives an arithmetic term for a sufficiently large T . The required majorizations are theorems of the arithmetic reduct. Boolean connectives and negation do not introduce additional predicate arguments.

Choose $c < E(T)$ that codes P below T . Replace each atomic subformula $P(t)$ in δ by $\text{Bit}(c, t)$, obtaining an arithmetic formula $\delta_c(z, \bar{a})$ in which P no longer occurs. An external induction on the finite syntax of δ now gives

$$(M, P) \models \delta(z, P, \bar{a}) \iff M \models \delta_c(z, \bar{a}) \quad (z \leq b + 1). \quad (6.7)$$

For atomic predicate formulas this is precisely the choice of c ; the quantified steps follow because all their bounded ranges were included in the construction of T .

The assumed induction premises for δ therefore imply

$$M \models \delta_c(0, \bar{a}) \quad \text{and} \quad M \models \forall z < b (\delta_c(z, \bar{a}) \rightarrow \delta_c(z + 1, \bar{a})).$$

Ordinary arithmetic induction, with $b, c, \bar{a}$ as parameters, gives $M \models \delta_c(b, \bar{a})$. This contradicts (6.7). Although E is named, this last induction is legitimate in the arithmetic reduct: the named function is a definitional extension of a function whose existence and uniqueness are provable in PA, so the arithmetic formula can be translated back into the original language. No induction instance containing P was used in this direction. $\square$

Remark 6.4. The theorem identifies bounded induction with the condition that every bounded restriction of P has an internal code. It does not identify this condition with full induction in the expanded language. The next theorem gives product measures under which these two properties differ almost surely.

6.3 An infinite locally finite predicate defines the standard cut

Call P *externally locally finite* if $P \cap [0, x)$ is a finite set in the ambient universe for every $x \in M$. The word “externally” is essential: internally finite sets in a nonstandard model can have infinitely many external elements.

Lemma 6.5 (Counting finite lists). *If M is nonstandard and $P \subseteq M$ is externally infinite and externally locally finite, then $\text{Cut}(P) = M$, but a $\Sigma_1(P)$ formula in the original arithmetic language defines exactly ω^M . In particular, (M, P) satisfies $\text{I}\Delta_0(P, E)$ and fails $\text{I}\Sigma_1(P)$.*

Proof. Every restriction $P \cap [0, x)$ is an externally finite set of elements of M . Such a set has a binary code in M , obtained by adding the corresponding finitely many powers of 2. Thus $\text{Cut}(P) = M$, and bounded induction follows from Theorem 6.3.

For the stronger obstruction, define the bounded formula

$$\begin{aligned} \Lambda(u, v, y, n, P) \equiv & v > 0 \wedge \forall i < n \exists r < y (\text{Beta}(u, v, i, r) \wedge P(r)) \\ & \wedge \forall i < n \forall j < n \forall r < y \forall s < y \\ & \left[(i < j \wedge \text{Beta}(u, v, i, r) \wedge \text{Beta}(u, v, j, s)) \rightarrow r < s \right]. \end{aligned} \quad (6.8)$$

Then

$$\ell(n, P) \equiv \exists u \exists v \exists y \Lambda(u, v, y, n, P) \quad (6.9)$$

is a $\Sigma_1(P)$ formula. It asserts that there is an internally coded strictly increasing n -term list of elements of P , all below one bound y .

For every standard n , the infinitude of P supplies n distinct elements; order them increasingly and encode this externally finite list by the Beta coding theorem. Thus $(M, P) \models \ell(n, P)$.

If n were nonstandard and u, v, y witnessed $\ell(n, P)$, then every standard index $i \in \omega^M$ would satisfy $i < n$. The associated remainders would therefore give infinitely many distinct elements of $P \cap [0, y)$, because the second clause of Λ makes them strictly increasing. This contradicts external local finiteness. Consequently ℓ defines exactly ω^M . Since M is nonstandard, this is a proper successor-closed subset containing 0, so the induction instance for ℓ fails. $\square$

6.4 Three regimes for independent sparse predicates

Let the coordinates be independent with $\mathbb{P}(P(a)) = p_a$, where $0 \leq p_a \leq \frac{1}{2}$. All sums in the following theorem are external sums of nonnegative real numbers. Put

$$W = \sum_{a \in M} p_a, \quad W_x = \sum_{a < x} p_a.$$

The zero-modal cut classification says $\text{Cut}(P) = S(p) = \{x \in M : W_x < \infty\}$ almost surely.

Theorem 6.6 (Three induction regimes). *For a countable nonstandard model M and the preceding independent product law, exactly one of the following deterministic alternatives holds, with the indicated conclusion almost surely.*

1. *If $W < \infty$, then P is externally finite and (M, P) satisfies full induction for every formula in the language with P .*
2. *If $W = \infty$ and $W_x < \infty$ for every $x \in M$, then P is externally infinite and externally locally finite. The expansion satisfies $\text{I}\Delta_0(P, E)$, but fails $\text{I}\Sigma_1(P)$.*
3. *If $W_x = \infty$ for some $x \in M$, then $\text{Cut}(P)$ is proper. The expansion fails both $\text{I}\Delta_0(P, E)$ and $\text{I}\Sigma_1(P)$.*

In particular, within this zero-modal product family, satisfaction of $\text{I}\Sigma_1(P)$ and satisfaction of full predicate induction each have probability 1 when $W < \infty$, and probability 0 when $W = \infty$.

Proof. If $W < \infty$, the first Borel–Cantelli lemma makes P externally finite almost surely. Write a realized finite predicate as $P = \{a_1, \dots, a_k\}$. Replace every occurrence of $P(t)$ in an arbitrary expanded-language formula by

$$t = a_1 \vee \dots \vee t = a_k$$

(or by a false formula when $k = 0$). This yields an arithmetic formula with parameters from M . The corresponding ordinary induction instance holds in M , proving full predicate induction.

For the second alternative, the first Borel–Cantelli lemma applied to each $[0, x)$ gives external local finiteness. There are only countably many x , so this conclusion holds simultaneously for all initial segments on one event of probability 1. The second Borel–Cantelli lemma, using independence and $W = \infty$, makes P externally infinite almost surely. Apply Lemma 6.5.

In the third alternative the zero-modal coding-cut theorem gives $\text{Cut}(P) = S(p) \neq M$ almost surely. Bounded induction fails by Theorem 6.3, and bare-language $\Sigma_1(P)$ induction fails by Corollary 6.2. These alternatives exhaust the possibilities for the nonnegative sums W and W_x . $\square$

Corollary 6.7 (Full support does not determine induction). *Each of the three regimes occurs for a product measure that assigns positive probability to every nonempty finite cylinder in 2^M .*

Proof. Enumerate M without repetitions as $(a_j)_{j \geq 0}$ and put $r_{a_j} = 2^{-j-4}$. Taking $p_a = r_a$ gives the first regime. For the second, choose a strictly increasing sequence (s_n) cofinal in M , set $p_{s_n} = 1/4$, and set $p_a = r_a$ elsewhere. Every bounded initial segment contains only finitely many s_n , so every W_x is finite, while $W = \infty$. Taking $p_a = 1/2$ everywhere gives the third regime, with coding cut ω^M almost surely. In all three constructions, $0 < p_a < 1$ at every coordinate, which gives full support. $\square$

Remark 6.8 (The standard-model exception). The nonstandard hypothesis in Theorem 6.6 is necessary. In the standard model every subset of $\mathbb{N}$ satisfies every semantic first-order induction instance, regardless of its probability law. An infinite locally finite predicate then makes $\ell(n, P)$ true for all n , so it defines no proper cut. The obstruction above comes from the coexistence of standardly finite lists and nonstandard possible list lengths.

7 Category, finite observations, and two orders of limits

For each endpoint let

$$A_a = \{P \in 2^M : a \in \text{Cut}(P)\}, \quad \mathcal{E}_I = \{P \in 2^M : \text{Cut}(P) = I\}.$$

The set $\mathcal{D}_a \subseteq 2^{[0, a)}$ of coded traces is countable. The global event A_a generally is not: the bits outside $[0, a)$ are arbitrary. This distinction is useful when discussing its topology and measure.

Proposition 7.1 (Category and Borel complexity). *For every nonstandard a , A_a is dense, meager, and F_σ , and its complement is dense. The exact-standard-cut fiber $\mathcal{E}_{\omega^M}$ is a dense G_δ . Every other exact-cut fiber is dense and meager. Each $\mathcal{E}_I$ is Borel, with the uniform upper bound Π_3^0 .*

Proof. For a fixed internal code c , its matching event below a is the intersection of the clopen conditions $P(t) = \text{Bit}(c, t)$ for $t < a$. It is closed. It has empty interior when a is nonstandard: a basic cylinder fixes only finitely many coordinates, so one can change an unfixed coordinate below a to disagree with the code. Thus A_a is a countable union of closed nowhere-dense sets.

To see density, fix any finite cylinder. Code precisely the finitely many ones it prescribes below a and use that code throughout the remaining part of $[0, a]$; outside the interval satisfy the cylinder arbitrarily. This produces a point of A_a . Its complement is dense by the Baire category theorem, since A_a is meager in a Cantor space.

Standard endpoints belong to every coding cut. Hence

$$\mathcal{E}_{\omega^M} = \bigcap_{a \in M \setminus \omega^M} (2^M \setminus A_a)$$

is a dense G_δ . For a cut $I \neq \omega^M$, choose a nonstandard $a \in I$; then $\mathcal{E}_I \subseteq A_a$, making the fiber meager. Its density follows from Theorem 4.2: it has measure one for a law of full support, so every nonempty open set meets it. Finally,

$$\mathcal{E}_I = \left(\bigcap_{a \in I} A_a \right) \cap \left(\bigcap_{a \notin I} (2^M \setminus A_a) \right). \quad (7.1)$$

The first intersection is a countable intersection of F_σ sets, and the second is G_δ . This gives Π_3^0 . $\square$

Thus category selects one coding cut, whereas full-support nonatomic probability laws can select any cut. Distinct cuts also give mutually singular realizing laws: their disjoint Borel fibers have full measure. There is no conflict between these statements. Full support requires positive mass on open sets; it does not require positive mass on all comeager sets.

Remark 7.2 (The complexity bound is not asserted optimal). If a cut I has the form $I = \{x : x < a + n \text{ for some standard } n\}$, then the condition $I \subseteq \text{Cut}(P)$ is equivalent to $a \in \text{Cut}(P)$, by successor closure. Its exact fiber is therefore the difference of two F_σ sets, $A_a \setminus \bigcup_{b \notin I} A_b$, and lies in Δ_3^0 . One must not claim that every nonstandard-cut fiber is Π_3^0 -complete. The exact complexity for general cuts is a separate research problem.

7.1 A finite observation never certifies noncoding

Fix a nonstandard a and enumerate $[0, a]$ externally as $t_0, t_1, \dots$. Enumerate all internal codes as $c_0, c_1, \dots$; repeated traces are harmless. For $K, N \geq 1$, define

$$u_{K,N}(P) = \mathbf{1}_{\{\exists j < K \ \forall i < N: P(t_i) = \text{Bit}(c_j, t_i)\}}. \quad (7.2)$$

Here K and N are external integers. Each test uses only finitely many predicate bits and finitely many proposed internal descriptions.

Theorem 7.3 (Exact noncommutation of finite tests). *For every predicate P ,*

$$\lim_{K \rightarrow \infty} \lim_{N \rightarrow \infty} u_{K,N}(P) = \mathbf{1}_{A_a}(P), \quad \lim_{N \rightarrow \infty} \lim_{K \rightarrow \infty} u_{K,N}(P) = 1. \quad (7.3)$$

In particular, under fair product measure the first limit is zero and the second is one almost surely.

Proof. For a fixed finite set of K codes, a code that does not match the full interval has a first disagreeing observed coordinate. If none of the K codes matches, after finitely many tests all are excluded. If one matches, every test accepts. Therefore the inner limit over N tests whether one of these K codes matches the whole interval. Letting K increase gives exactly A_a .

For a fixed N , all prescribed bits on $\{t_0, \dots, t_{N-1}\}$ have an internal code, by Lemma 2.2. That code eventually appears in the enumeration, so the inner limit over K is one. The fair-measure assertion follows from Corollary 3.2. $\square$

This is a precise form of the obstacle to inferring an infinite coding claim from unrestricted finite checks. Every finite partial observation extends both to a coded trace and to a noncoded trace. Adaptively choosing the next coordinate does not remove the obstacle: a completed finite transcript still fixes only finitely many coordinates. Finite data can support probabilistic conclusions under a specified law, but it cannot give a logically certain noncoding certificate for an arbitrary predicate.

Proposition 7.4 (The coding-cut map has exact Baire class two). *For nonstandard M , the map $P \mapsto \text{Cut}(P)$, viewed as a map $2^M \rightarrow 2^M$ of characteristic functions, is Baire class two, is not Baire class one, and is discontinuous at every point.*

Proof. Choose an exhaustion F_N of all of M by finite sets. Define $g_{K,N}(P)(a)$ to mean that one of the first K codes agrees with P on $F_N \cap [0, a)$. Each coordinate depends on finitely many bits, so $g_{K,N} : 2^M \rightarrow 2^M$ is continuous. For fixed K , its pointwise limit as $N \rightarrow \infty$ records matching one of the first K codes on the whole interval, by the finite-code argument in Theorem 7.3. Call this limit g_K ; it is Baire class one. Letting $K \rightarrow \infty$ gives the coding-cut map, hence Baire class two.

Fix any nonstandard endpoint a . Its coordinate is the indicator of A_a , whose set and complement are both dense. That coordinate is discontinuous everywhere. A Baire-class-one map from a Polish space to a metric space has points of continuity on a dense G_δ ; equivalently one can apply the usual Baire theorem to this discrete-valued coordinate. It cannot be Baire class one. A continuous coding-cut map at any point would make every coordinate continuous there, which is impossible. $\square$

7.2 Stability under locally summable perturbations

The weak instability in Corollary 4.3 has a useful counterpart. A perturbation that produces only finitely many changes on each bounded interval cannot alter a coding cut.

Proposition 7.5 (Local error stability). *Suppose a coupling of two random predicates P, Q satisfies*

$$\sum_{t < a} \mathbb{P}(P(t) \neq Q(t)) < \infty \quad \text{for every } a \in M.$$

Then $\text{Cut}(P) = \text{Cut}(Q)$ almost surely. No independence is needed. In particular, if two independent product bias families p, q satisfy

$$\sum_{t < a} |p_t - q_t| < \infty \quad (a \in M),$$

their deterministic almost-sure coding cuts are the same.

Proof. The union-tail argument from Theorem 3.1 gives only finitely many mismatches below each endpoint, simultaneously by countability of M . Use finite-change invariance. For product laws, take

mutually independent uniform random variables U_t and set $P(t) = \mathbf{1}_{\{U_t < p_t\}}$, $Q(t) = \mathbf{1}_{\{U_t < q_t\}}$. Their mismatch probability is $|p_t - q_t|$, and each marginal process is the desired independent product law. $\square$

The theorem identifies a concrete notion of perturbation under which the cut is stable, even though weak convergence alone does not preserve it.

8 How much independence is actually necessary?

The product proof used exponentially small matching probabilities. For the infinite coding conclusion it is enough that the matching probability of each fixed code tends to zero. This permits a sharp reduction in the independence assumption for a single predicate.

Theorem 8.1 (Pairwise independence suffices for the full bias formula). *Let the bits $(P(t))_{t \in M}$ be pairwise independent, with arbitrary marginal probabilities $p_t \in [0, 1]$. Define b, ε as in (3.2). Then*

$$\text{Cut}(P) = \text{Cut}(b) \cap S(\varepsilon) \quad \text{almost surely.} \quad (8.1)$$

Proof. The summable case in Theorem 3.1 used no independence and therefore remains unchanged. In the divergent case, fix an internal candidate pattern d below a and write $Y_t = \mathbf{1}_{\{P(t) \neq d(t)\}}$. Coordinatewise transformations preserve pairwise independence. Their probabilities satisfy $q_t = \mathbb{P}(Y_t = 1) \geq \varepsilon_t$.

For a finite set $F \subseteq [0, a)$, put $Z_F = \sum_{t \in F} Y_t$ and $m_F = \sum_{t \in F} q_t$. Then

$$\text{Var}(Z_F) = \sum_{t \in F} q_t(1 - q_t) \leq m_F.$$

The elementary second-moment inequality $\mathbb{P}(Z_F > 0) \geq (\mathbb{E}Z_F)^2 / \mathbb{E}Z_F^2$, obtained by Cauchy–Schwarz, gives for $m_F > 0$

$$\mathbb{P}(Z_F = 0) \leq \frac{\text{Var}(Z_F)}{m_F^2 + \text{Var}(Z_F)} \leq \frac{1}{1 + m_F}. \quad (8.2)$$

As F exhausts the interval, $m_F \rightarrow \infty$. Thus matching d on the whole interval has probability zero. Countability of the codes and endpoints completes the proof just as before. $\square$

In particular, pairwise-independent fair bits reveal the standard cut. The finite version has a sharp parity effect, not the exponential bound available under full independence.

Theorem 8.2 (Sharp finite atom bound; classical moment extremum). *Among all distributions of $N \geq 1$ pairwise-independent fair bits, the largest possible probability of any specified word is*

$$q_N = \begin{cases} 1/(N+1), & N \text{ odd,} \\ 1/(N+2), & N \text{ even.} \end{cases} \quad (8.3)$$

Consequently the probability of any catalog of K words is at most $\min(1, Kq_N)$; this last catalog bound is not asserted sharp for arbitrary K .

The atom formula is a special case of the known moment bounds presented by Benamini, Gurel-Gurevich, and Peled, Proposition 25, drawing on Boros–Prékopa [5, 6]. The short proof below also provides explicit extremizers for the finite problem and makes the infinite application transparent.

Proof. Complement coordinates as needed to make the target the zero word; this preserves pairwise-independent fairness. Let W be the number of ones. Then $\mathbb{E}W = N/2$ and $\text{Var}(W) = N/4$.

If $N = 2a - 1$, the nonnegative polynomial $(W - a)^2/a^2$ is one at $W = 0$. Its expectation is $(a/2)/a^2 = 1/(2a)$, proving the odd bound. Equality is attained by putting probability $1/(2a)$ on the zero word and the remaining probability uniformly on all words of weight a .

If $N = 2a$, the polynomial $(W - a)(W - a - 1)/(a(a + 1))$ is nonnegative at every integer W and equals one at $W = 0$. Its expectation is $(a/2)/(a(a + 1)) = 1/(2a + 2)$. Equality is attained by distributing mass uniformly within each of the following layers:

$$\mathbb{P}(W = 0) = \frac{1}{2a + 2}, \quad \mathbb{P}(W = a) = \frac{1}{2}, \quad \mathbb{P}(W = a + 1) = \frac{a}{2a + 2}.$$

To verify both constructions, their first two factorial moments are $\mathbb{E}W = N/2$ and $\mathbb{E}[W(W - 1)] = N(N - 1)/4$. Uniformity within layers therefore gives $\mathbb{P}(X_i = 1) = 1/2$ and $\mathbb{P}(X_i = X_j = 1) = 1/4$ for distinct i, j , hence all four pairwise joint probabilities are $1/4$. For $N = 1$ the pairwise condition is vacuous and the stated fair construction still applies. The catalog bound is the union bound over its words. $\square$

Theorem 8.3 (Marginal fairness alone allows every cut). *For every cut I , there is a full-support nonatomic law on 2^M under which each individual bit is fair and $\text{Cut}(P) = I$ almost surely. Thus pairwise independence is the least level in the hierarchy of k -wise independent fair laws that forces the standard cut.*

Proof. Take a predicate X with the law in Theorem 4.2 for I , and an independent fair bit Z . Set $P(t) = X(t) \oplus Z$ for every t . Each $P(t)$ is fair. Its full law is the equal mixture of the law of X and its complement. Both components have full support and no atoms, so their mixture has the same properties. Complementation preserves the coding cut, giving $\text{Cut}(P) = I$. The pairwise-positive result is Theorem 8.1; choosing $I \neq \omega^M$ proves the claimed sharpness among the levels $k = 1, 2, \dots$ $\square$

8.1 Limits of the pairwise extension

Theorem 8.1 concerns a single predicate. It does not permit replacing joint independence by pairwise independence of all the individual bits in the Boolean-spectrum theorem. For example, let A_t be independent fair bits and let Z be a separate fair bit. Set

$$P_1(t) = A_t, \quad P_2(t) = A_t \oplus Z.$$

Every two distinct bits of this whole array are independent and fair, and each row has standard coding cut almost surely. Yet $P_1(t) \oplus P_2(t) = Z$ is constant in t and has coding cut M . Boolean operations can uncover higher-order dependence.

Even independence of the random subsets as whole rows is insufficient if their coordinates are not independent. Let P_1 be a uniformly random residue class modulo a fixed standard $q \geq 6$, and let P_2 be an independent iid Bernoulli($1/8$) predicate. The first row always has cut M , the second has cut ω^M , and their intersection has cut ω^M : on the selected residue class inside every nonstandard interval it still has infinitely many independent nondegenerate bits. All individual marginals are small. These examples explain why the Boolean theorem states independence of the whole array indexed by both row and position.

8.2 A general dependent-law criterion

Proposition 8.4 (Local atoms are the exact obstruction). *For an arbitrary Borel law ν on 2^M , let ν_a be its restriction law on $2^{[0,a]}$. Then*

$$\nu\{\text{Cut}(P) = \omega^M\} = 1 \iff \nu_a(\{d\}) = 0 \text{ for every nonstandard } a \text{ and every } d \in \mathcal{D}_a.$$

In particular, nonatomicity of every nonstandard restriction law is sufficient. Nonatomicity of the full law alone is insufficient.

Proof. The event $a \in \text{Cut}(P)$ is the union of the singleton-pattern events indexed by the countable family $\mathcal{D}_a$. Such a union is null exactly when all of its members are null. Now take the countable union over nonstandard endpoints. The final warning follows already from Theorem 4.2 with $I = M$. $\square$

Another sufficient condition allows substantial dependence. Suppose the coordinates are enumerated as $t_0, t_1, \dots$ and there are deterministic $\delta_n \in [0, 1/2]$ such that almost surely

$$\delta_n \leq \mathbb{P}(P(t_n) = 1 \mid P(t_0), \dots, P(t_{n-1})) \leq 1 - \delta_n.$$

For any finite selected collection of indices, successively conditioning on the complete preceding history bounds the probability of matching a fixed word there by $\prod(1 - \delta_n)$. Therefore

$$\text{Cut}(P) \subseteq \left\{ a : \sum_{n: t_n < a} \delta_n < \infty \right\} \quad \text{almost surely.}$$

This is an obstruction criterion, not an equality claim: correlations can still change what happens where the lower bounds are summable.

9 The finite candidate-budget problem

The unrestricted finite version of binary coding is vacuous: every binary word of ordinary finite length has an arithmetic code. A meaningful finite problem must therefore impose a budget on the candidate descriptions. Let $X = (X_1, \dots, X_N)$ have independent coordinates, with $\Pr(X_i = 1) = p_i$. Define

$$Q_N(K) = \max_{\substack{\mathcal{C} \subseteq \{0,1\}^N \\ |\mathcal{C}| \leq K}} \Pr(X \in \mathcal{C}), \quad 1 \leq K \leq 2^N.$$

The problem asks how many candidate codes are needed to capture a specified probability of the random predicate. This is precisely the almost-lossless fixed-length source-coding problem, expressed in terms of a candidate list; the information-theoretic optimization and its fixed-bias normal approximation are classical [3, 4]. Our purpose is to connect them explicitly with the summability criterion in the infinite problem, including a sparse regime where the finite budget has a polynomial staircase.

9.1 Exact optimization and the infinite limit

Choose a modal word b coordinate by coordinate, and put $\varepsilon_i = \min(p_i, 1 - p_i)$. Thus the indicators of disagreement with b are independent Bernoulli variables of parameters ε_i . Write $a_N = \prod_{i=1}^N (1 - \varepsilon_i)$.

Proposition 9.1 (Exact finite optimization). *The value $Q_N(K)$ is the sum of the K largest atom probabilities of the product distribution. In particular,*

$$Q_N(K) \leq \min\{1, K a_N\}.$$

If $0 < \varepsilon_i < 1/2$, a word obtained by changing b on a set F has probability

$$a_N \prod_{i \in F} \frac{\varepsilon_i}{1 - \varepsilon_i}.$$

Consequently the optimal list consists of the K subsets F of smallest cost $\sum_{i \in F} \log((1 - \varepsilon_i)/\varepsilon_i)$, with arbitrary tie-breaking.

Proof. If a proposed list contains an atom of probability u while omitting an atom of larger probability v , exchanging the two increases its captured mass by $v - u$. Repeated exchanges prove the optimization statement. The modal atom has the greatest probability, namely a_N , giving the upper bound. The displayed product is the quotient of the probability of the changed word by the modal atom probability. $\square$

Here and below, a fixed integer K is also allowed when $K > 2^N$, by setting $Q_N(K) = 1$. For an infinite sequence of biases this convention makes the following statement uniform in N .

Theorem 9.2 (Finite budgets characterize the atomic regime). *For independent coordinates with $0 \leq \varepsilon_i \leq 1/2$, the following are equivalent:*

1. $\sum_{i=1}^{\infty} \varepsilon_i < \infty$;
2. for every $\eta > 0$ there is a finite K such that $Q_N(K) \geq 1 - \eta$ for every N ;
3. the infinite product distribution is concentrated on a countable set of words.

More precisely, if the sum diverges, $Q_N(K) \rightarrow 0$ for every fixed K . If it converges, the distribution is concentrated on the finite modifications of b , and

$$Q_N(K) \downarrow Q_{\infty}(K),$$

where $Q_{\infty}(K)$ is the sum of its K largest infinite-word atom masses. For every L and K one has the quantitative bound

$$0 \leq Q_L(K) - Q_{\infty}(K) \leq \sum_{i > L} \varepsilon_i.$$

In particular, if $\sum_{i > L} \varepsilon_i \leq \eta$, the single finite list consisting of all 2^L modifications of b on the first L coordinates captures probability at least $1 - \eta$, simultaneously at every truncation length.

Proof. Projection of any K words of length $N+1$ gives at most K words of length N , so $Q_{N+1}(K) \leq Q_N(K)$. Conversely, extend the K most probable length- L words by the modal suffix. Independence gives

$$Q_N(K) \geq Q_L(K) \prod_{L < i \leq N} (1 - \varepsilon_i), \quad N \geq L.$$

Together with the elementary union bound $1 - \prod(1 - \varepsilon_i) \leq \sum \varepsilon_i$, this proves

$$0 \leq Q_L(K) - Q_N(K) \leq \sum_{L < i \leq N} \varepsilon_i.$$

If $\sum \varepsilon_i = \infty$, then $a_N \leq \exp(-\sum_{i \leq N} \varepsilon_i) \rightarrow 0$; hence Proposition 9.1 gives $Q_N(K) \rightarrow 0$. Every infinite word is then an atom of mass at most $\lim_N a_N = 0$, so no countable set can have positive probability.

Suppose instead that $\sum \varepsilon_i < \infty$. With probability one there are only finitely many disagreements with b : indeed the probability of any disagreement after L is at most $\sum_{i > L} \varepsilon_i \rightarrow 0$. Thus the distribution is carried by the countable family of finite modifications. Those involving only the first L coordinates have probability at least $1 - \sum_{i > L} \varepsilon_i$, proving the bounded-budget assertion. Their projections give the simultaneous finite-list claim.

It remains to identify the limit. Let s_K be the sum of the K largest atom masses. Selecting their prefixes proves $Q_N(K) \geq s_K$. Given $\delta > 0$, choose a finite set F of atoms of total mass at least $1 - \delta$. For all sufficiently large N , the atoms in F have pairwise distinct length- N prefixes. A K -word list then captures mass at most s_K from F and at most δ from its complement. Therefore $\lim_N Q_N(K) \leq s_K + \delta$, and $\delta \downarrow 0$ gives the claimed equality. Taking $N \rightarrow \infty$ in the preceding quantitative bound finishes the proof. $\square$

Remark 9.3 (Summable minority mass is not summable entropy). Let $\varepsilon_i = c/[i(\log i)^2]$ for $i \geq 3$, with a positive constant c small enough that $\varepsilon_i < 1/2$. Then

$$\sum_i \varepsilon_i < \infty, \quad \sum_i h(\varepsilon_i) = \infty, \quad h(t) = -t \log t - (1-t) \log(1-t).$$

The second assertion follows from $h(\varepsilon_i) \geq \varepsilon_i \log(1/\varepsilon_i) \geq c'/(i \log i)$ for sufficiently large i . Thus a countably supported product law may have infinite Shannon entropy. High-confidence finite candidate budgets nevertheless remain uniformly bounded by Theorem 9.2. Shannon-entropy summability cannot replace minority-mass summability in the infinite coding theorem.

9.2 An exact Hamming-layer formula

Assume now that every disagreement probability is the same $p \in (0, 1/2)$. Put

$$B_N(r) = \sum_{j=0}^r \binom{N}{j}, \quad B_N(-1) = 0.$$

Given K , there is a unique $r \in \{0, \dots, N\}$ such that $B_N(r-1) < K \leq B_N(r)$. Define

$$\theta = \frac{K - B_N(r-1)}{\binom{N}{r}} \in (0, 1].$$

Theorem 9.4 (Exact biased finite answer). *If $W_N \sim \text{Bin}(N, p)$, then*

$$Q_N(K) = \Pr(W_N < r) + \theta \Pr(W_N = r).$$

An optimal list contains every word with fewer than r disagreements from the mode and any $K - B_N(r-1)$ words having exactly r disagreements. For $p = 1/2$ the answer is instead simply $Q_N(K) = K/2^N$.

Proof. A specified word of weight j has probability $p^j(1-p)^{N-j}$. The ratio between probabilities at consecutive weights is $p/(1-p) < 1$, so the K largest atoms consist of complete successive Hamming layers and at most one partial layer. Each atom in the partial layer has the same probability, so the fraction of that layer's probability captured is exactly θ . Uniform bits give equal probabilities 2^{-N} . $\square$

9.3 The fixed-bias critical window

Write

$$h(p) = -p \log p - (1-p) \log(1-p), \quad v(p) = p(1-p) \left(\log \frac{1-p}{p} \right)^2.$$

Let $K_N^*(\tau)$ be the least integer K for which $Q_N(K) \geq \tau$, where $0 < \tau < 1$ is fixed. The following is the Bernoulli case of the classical third-order source-coding expansion [3, 4]. An elementary proof is included to make explicit which part of the finite problem is solved.

Theorem 9.5 (Normal transition and third-order budget). *For fixed $0 < p < 1/2$ and $0 < \tau < 1$,*

$$\log K_N^*(\tau) = Nh(p) + \sqrt{Nv(p)} \Phi^{-1}(\tau) - \frac{1}{2} \log N + O_{p,\tau}(1).$$

Equivalently, for every fixed real s , if

$$\log K_N = Nh(p) + \sqrt{Nv(p)} s - \frac{1}{2} \log N + o(\sqrt{N}),$$

then $Q_N(K_N) \rightarrow \Phi(s)$. In particular, a budget of exponential rate $R < h(p)$ captures probability tending to zero, whereas rate $R > h(p)$ captures probability tending to one (with K_N capped at 2^N).

Proof. Let r_N be the least integer satisfying $\Pr(W_N \leq r_N) \geq \tau$. The Berry–Esseen inequality for a binomial sum, together with positivity of the normal density at $z = \Phi^{-1}(\tau)$, gives

$$r_N = Np + \sqrt{Np(1-p)} z + O_{p,\tau}(1).$$

For completeness, the cumulative binomial distribution differs from the normal distribution evaluated at $(r - Np)/\sqrt{Np(1-p)}$ by $O(N^{-1/2})$. Moving the normalized argument by a sufficiently large constant times $N^{-1/2}$ brackets τ on its two sides, establishing the asserted $O(1)$ quantile error.

The exact layer formula gives

$$B_N(r_N - 1) < K_N^*(\tau) \leq B_N(r_N).$$

When $r_N/N \rightarrow p < 1/2$, both boundary sums are within constant factors of $\binom{N}{r_N}$. To see the upper bound, consecutive terms read backwards have ratio at most $r_N/(N - r_N + 1)$, bounded above by a constant less than one. For the lower bound use the last term $\binom{N}{r_N-1} = \binom{N}{r_N} r_N/(N - r_N + 1)$, whose ratio is bounded below by a positive constant. Hence

$$\log K_N^*(\tau) = \log \binom{N}{r_N} + O(1).$$

Uniform Stirling estimates in a fixed neighborhood of p give

$$\log \binom{N}{r} = Nh(r/N) - \frac{1}{2} \log N + O(1).$$

Taylor expansion at p , using $r_N - Np = O(\sqrt{N})$ and $h'(p) = \log((1-p)/p)$, yields the claimed expansion.

For the transition statement, choose any $\delta > 0$. The expansion just proved, applied to $\tau_- = \Phi(s - \delta)$ and $\tau_+ = \Phi(s + \delta)$, implies $K_N^*(\tau_-) < K_N < K_N^*(\tau_+)$ for sufficiently large N . Monotonicity gives $\Phi(s - \delta) \leq \liminf Q_N(K_N) \leq \limsup Q_N(K_N) \leq \Phi(s + \delta)$. Let $\delta \downarrow 0$. The exponential-rate assertion follows in the same way from fixed confidence levels approaching zero or one. $\square$

9.4 A sparse-noise Poisson staircase

The normal transition is not uniform as $p \downarrow 0$. If the expected number of disagreements stays finite, the candidate-budget scale changes from exponential to polynomial.

Theorem 9.6 (Sparse candidate-budget staircase). *Let $p_N = \lambda/N$ with fixed $\lambda > 0$. Fix an integer $r \geq 1$ and a constant $\kappa > 0$, and let $K_N/N^r \rightarrow \kappa$. For $Z \sim \text{Poisson}(\lambda)$,*

$$\lim_{N \rightarrow \infty} Q_N(K_N) = \Pr(Z < r) + \min\{\kappa r!, 1\} \Pr(Z = r).$$

For a fixed positive integer budget K , the limiting value is $e^{-\lambda}$. More generally, for an integer $r \geq 0$ and a fixed $r < \gamma < r + 1$, if $K_N = N^{\gamma+o(1)}$, then $Q_N(K_N) \rightarrow \Pr(Z \leq r)$.

Proof. For fixed r , Stirling's finite-product form gives

$$B_N(r-1) = O(N^{r-1}), \quad \binom{N}{r} = \frac{N^r}{r!} (1 + O(N^{-1})), \quad \binom{N}{r+1} \asymp N^{r+1}.$$

If $\kappa r! < 1$, the optimal list eventually consists of every layer below r and a fraction $\theta_N \rightarrow \kappa r!$ of layer r . If $\kappa r! > 1$, it contains all layers through r and a fraction tending to zero of layer $r + 1$. At equality, either description can occur along subsequences, but the two limiting answers coincide. For every fixed j ,

$$\Pr\{\text{Bin}(N, \lambda/N) = j\} = \binom{N}{j} (\lambda/N)^j (1 - \lambda/N)^{N-j} \longrightarrow e^{-\lambda} \frac{\lambda^j}{j!}.$$

The exact layer formula now gives the asserted limit. A fixed positive budget contains the modal word and only finitely many atoms from weight one, each of probability tending to zero, yielding $e^{-\lambda}$. If $N^r \ll K_N \ll N^{r+1}$, every layer through r is included and a vanishing fraction of layer $r + 1$ is included, proving the last assertion. $\square$

For example, when $\lambda = 1$, a fixed budget captures asymptotic mass e^{-1} , a budget $K_N \sim \kappa N$ captures $e^{-1}(1 + \min\{\kappa, 1\})$, and a budget $K_N \sim \kappa N^2$ captures $e^{-1}(2 + \frac{1}{2} \min\{2\kappa, 1\})$. The step locations record how many minority errors the candidate list can name. This is the finite counterpart of the fact that summable independent noise makes only finitely many changes to the modal word.

Remark 9.7 (A fixed infinite law and a triangular array are different). Theorem 9.2 concerns prefixes of one fixed infinite sequence of biases. The sparse theorem uses a different product law at each length N , with $p_N = \lambda/N$ at all of its sites. Bounded total row mass in that triangular array does not imply uniformly bounded high-confidence catalogs: for confidence above $e^{-\lambda}$, the budgets grow. There is no contradiction with the summability theorem because no single infinite bias sequence has those rows as all of its prefixes.

9.5 Why finite observations alone cannot certify noncoding

There is a crucial limitation when transferring these finite budgets back to a nonstandard arithmetic model. If x is nonstandard and $\mathcal{A}_x$ is the external countable family of internally coded binary words on $[0, x)$, then $\mathcal{A}_x$ realizes every prescribed pattern on every externally finite subset of $[0, x)$. Indeed one may code the finitely many coordinates whose prescribed values are one and give value zero

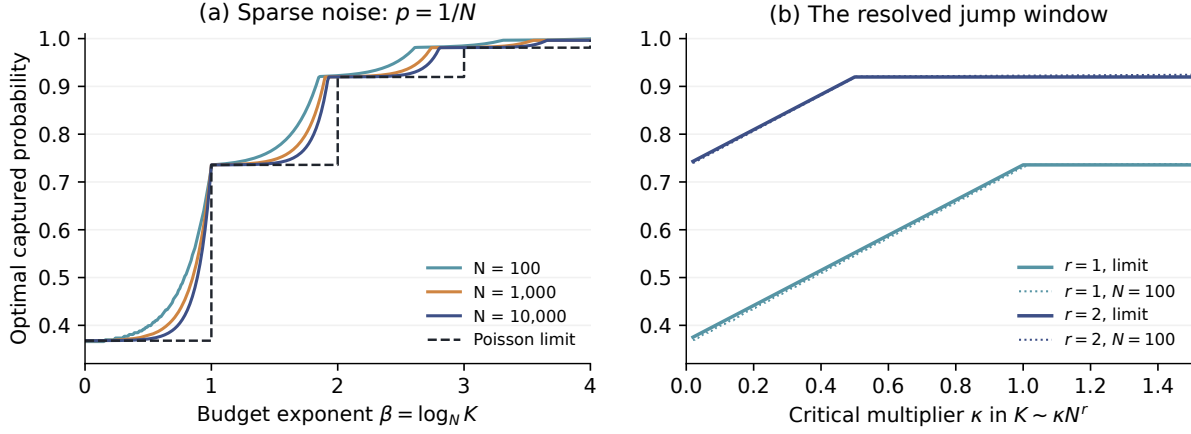


Figure 1: Exact finite optimal-catalog probabilities for sparse Bernoulli noise with $p = 1/N$. Left: the global polynomial-budget staircase; the dashed curve is the Poisson limit. Right: the resolved critical multiplier at $K \sim \kappa N^r$. Every curve is evaluated from the Hamming-layer formula; no simulation of nonstandard arithmetic is involved.

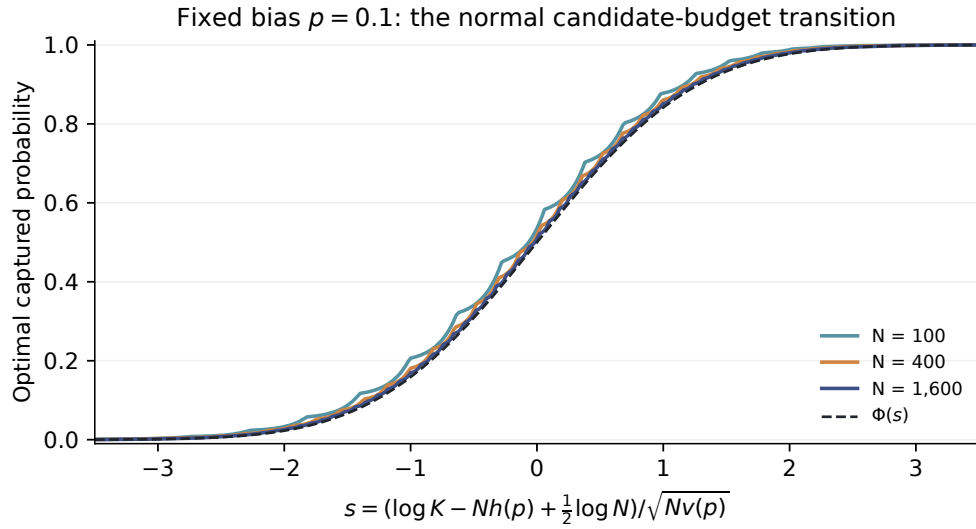


Figure 2: The fixed-bias normal transition. The standardized catalog budget uses the entropy term, the square-root dispersion term, and the $-\frac{1}{2} \log N$ correction in Theorem 9.5. Finite probabilities are computed from exact layer counts and numerical binomial probabilities.

elsewhere. Thus $\mathcal{A}_x$ is dense in the external product topology, even though in the nonsummable regime it has measure zero. No finite observation, including an adaptively selected finite set of coordinates, can logically certify that a word is not internally coded. The finite candidate-budget problem is therefore an additional, explicit resource restriction; its parameter K must not be silently identified with the full internal family of codes.

10 Consequences linking the constructions

The universal construction has a stronger logical feature than arbitrary coding cuts alone. Its Boolean outputs can be locally coded without satisfying the stronger induction scheme.

Corollary 10.1 (All nonconstant Boolean outputs fail stronger induction). *For a nonstandard M , in the realization of Theorem 5.2, and also in its vanishing-bias refinement, almost surely every nonconstant finitary Boolean observable $R = f(P)$ fails $\mathbb{I}\Sigma_1(R)$. If its coding cut is M , it nevertheless satisfies $\mathbb{I}\Delta_0(R, E)$.*

Proof. If the output cut is proper, use the bare-language coding formula from Corollary 6.2. If its cut is M , consider the deviation predicate D from the all-zero output value $f(\mathbf{0})$. The proof of Theorem 5.5 identifies its coding cut with the summability cut of its deviation probabilities. Therefore every bounded restriction of D is externally finite almost surely. The marker construction makes D infinite almost surely, including in the vanishing-bias refinement. Lemma 6.5 gives failure of $\mathbb{I}\Sigma_1(D)$. If R is the complement of D , literal substitution of $\neg R$ for D preserves the formula class and transfers the failed induction instance. Complete local coding gives the bounded induction statement. Countably many observables allow simultaneous conclusions. $\square$

10.1 What the finite probabilities say about internal codes

The finite optimization and the infinite classification use the same mechanism at different quantifier scales. With independent coordinates, the largest atom on a finite set F is

$$\prod_{t \in F} (1 - \varepsilon_t).$$

If the accumulated minority mass diverges, this tends to zero. A fixed finite catalog then loses all captured probability, and each individual infinite word has probability zero. Countability of the internal catalog is what turns that last statement into almost-sure noncoding.

When minority mass is summable, the observed infinite word is a finite modification of its mode. A finite catalog can capture arbitrarily high confidence uniformly in the observation length. Internal codability is then determined by the modal word. These statements explain why the infinite theorem is about minority-mass summability, whereas the exponential growth rate of finite optimal catalogs is Shannon entropy. The two quantities answer different questions and need not agree.

At sparse noise $p_N = \lambda/N$, the finite experiment retains a bounded expected number of errors while their possible positions proliferate. Naming all patterns with r errors costs order N^r . That count produces the Poisson staircase. It is a finite positional cost, even though the number of errors has a tight limiting distribution.

For reference, the following values are obtained directly from the exact finite formula at $N = 10,000$ and $p = 1/N$. They are numerical evaluations, not experimental estimates from random trials.

Catalog size	Finite captured probability	Limiting probability
$K = N/4$	0.45979872	$1.25/e \approx 0.45984930$
$K = N$	0.73572209	$2/e \approx 0.73575888$
$K = N^2/2$	0.91968940	$2.5/e \approx 0.91969860$

11 A proposed formalization route in ProveIt

[write] **A proposal, not repository status; 6 October 2026.** The delivered title of this section is “A concrete formalization route in ProveIt”. It is the source’s proposal. None of the modules in the table below exists in the repository at HEAD (3988bf5c4), and no statement of this report is verified. What the repository does contain is listed by declaration in Section 1.3: the beta-coding lemmas `finite_list_beta_code` and `finite_vector_beta_code` cover externally finite lists only (the binary relation `Bit` and the internal coding lemmas of the first row of the table are not formalized), and `not_sat_inductionForm` concerns the formula `cutAt evalFormula`, not the coding formulas of Section 6. The plan is carried as Question Q13 of Section 13.1.

The mathematics naturally separates into interfaces that can be verified independently and then connected. An efficient first target is the single-predicate classification and its pairwise-independent extension; the Boolean universality theorem can follow once the summability and marker interfaces are available.

11.1 Use raw first-order arithmetic semantics

The inspected `NotFinitelyAxiomatizable` project explicitly works with raw first-order arithmetic structures satisfying the PA schema [8]. That distinction is essential. A host-level model interface whose induction field ranges over *all* ambient predicates would exclude nonstandard models and would not express the setting here.

Its `FiniteBetaCoding.lean` proves that externally finite lists of arbitrary elements of a raw PA model can be packed into beta codes [9]. This addresses the important case in which a changed bit position or a selected list entry is nonstandard. Its `EvaluatorCutContract.lean` isolates an existing definable-cut induction argument [10]; the present random coding predicate needs a new contract, rather than an invocation of that theorem with its evaluator hypotheses omitted.

The separate `ListCoding` README states a more limited formal boundary for many exported formulas: their evaluation specifications are metatheorems for the standard natural-number interpretation, not automatic internal PA derivations of every universal closure [7]. Those results are useful starting points, but a theorem about arbitrary nonstandard models must pass through the appropriate syntactic theorem and soundness interfaces. The present report does not silently transfer standard-model correctness to every model.

11.2 Suggested proof modules

Module or interface	Required statement and verification boundary
Internal binary coding	Define the bit formula and prove restriction, successor extension, Boolean closure, and externally finite modification in arbitrary raw PA models. Record which facts are internal theorems and which are external finite constructions.
Abstract coded traces	Package a countable family of patterns on each initial interval, closed under finite changes and restriction. Prove the cut and filtration lemmas without probability.
External summability	Use nonnegative-real sums or finite-subset suprema. Prove successor closure, invariance under summable perturbations, and the marker characterization. Do not substitute an internal arithmetic sum.
Probability classification	Prove the finite-tail implication, nullity of a fixed trace, and the countable-code union. For pairwise independence, verify the variance and second-moment bound separately.
Countable spectra	Formalize the enumeration of constraints and the fresh-marker recursion. The decisive invariant controls all labels containing a fixed finite set, not only one cofinal sequence.
Boolean observations	Compute minimal sensitive supports from a finite truth table. Prove the disjoint-event lower bound and the union upper bound before connecting their summability cuts.
Induction syntax	Construct the explicit bounded formula with E and the bare-language beta formulas. Verify query bounds by recursion on syntax; use arithmetic induction only after eliminating P locally.
Finite catalogs	Certify the exchange argument, binomial-layer counts, and quadratic dual bounds using exact rational arithmetic. The existing Python checks are evidence for examples, not trusted proof objects.

A formalization should expose theorem dependencies and avoid stating the probabilistic conclusion as an axiom. The desired end product is an actual proof that almost every predicate has the asserted coding cut, with the countability assumptions visible in its type. Full formalization was not performed in this report.

12 Verification and source audit

12.1 Mathematical checks

Independent derivations agreed on the arbitrary-bias formula, arbitrary-cut realization, and the countable marker construction. A separate skeptical review checked the Boolean comparisons, the weak-limit statement, the beta-code conversion, and the induction phase diagram. The review produced counterexamples to stronger but false formulations involving only marginal fairness, only pairwise independence of array entries, independence of rows without coordinate independence, or probabilities allowed to approach one. The article states the narrower hypotheses that withstand these checks. This is an internal research audit, not external peer review.

12.2 Executed finite checks

The included script `verify_finite_budgets.py` reproduces the figures and CSV data and performs the following tests. The exact checks use rational arithmetic and a second implementation that enumerates the distribution’s atoms.

Check	Count
Exact iid catalog formula versus sorted atom sums, $N \leq 10$	6,138
Nonidentical-product projection and modal-extension bounds	494
Floating evaluation versus exact rational answers	114
Complete pairwise-independent extremizing distributions, $1 \leq N \leq 10$	10
Exact fair-marginal identities in those distributions	55
Exact pairwise joint-probability identities	660
Pointwise quadratic majorant inequalities	65
Exact dual-expectation identities	10

All checks passed. The maximum absolute error in the floating spot checks was 3.33×10^{-16} . The generated JSON records the runtime versions, counts, and scope. No finite structure was presented as a simulation of a nonstandard model. The infinite results rest on the mathematical proofs, and the tests establish neither those theorems nor historical priority.

12.3 Source boundary

Repository inspection used the pinned revision `3b9458b31f459b480ab43a068d7cc56f4cfdbb28`. The focused files are listed in the bibliography and in the package’s source-audit text. Searches for random predicates and coin flipping did not return an existing repository treatment of the precise problem selected here. This focused search is not an exhaustive audit of the entire repository or its incoming archives.

The public Glazer source used for the direct research connection is the official seminar abstract, not an inferred private preference. The underlying video was not used as evidence for a particular formula. The arithmetic-randomness theorem announced there, the established source-coding asymptotics, and the known finite pairwise moment optimum are explicitly credited. Our results are formulated so that their proofs can be assessed independently of any priority claim.

[write] 6 October 2026. At the write’s HEAD (`3988bf5c4`) the four cited repository files are byte-identical to the pin, and the search statement above is still true (Section 1.3, “Stale claims”). The finite checks were rerun at placement and at the write on copies of the delivered script (Python 3.13.5, NumPy 2.5.3, SciPy 1.18.1, Matplotlib 3.11.2; 25–35 s). Every count of the table above and the maximal error 3.33×10^{-16} were reproduced. The regenerated JSON differs from the shipped one only in its version block; the four CSV tables have the same rows, and differ in 19, 3, 7 and 3 rows by floating-point noise of at most 1.02×10^{-14} . In the repository the script is `code/verify_finite_budgets.py` and its JSON is `data/finite_budget_verification.json`; because the script writes its outputs beside itself, it must be rerun on a copy (see the README).

13 Further research questions and topics

The following are proposed problems arising from this report. They are not all asserted to be previously unposed problems in the literature. Each entry identifies a concrete next target and the

part of the present work that makes it accessible.

- Q1. The weakest arithmetic base.** Determine the weakest arithmetic theory supporting the arbitrary-bias classification and the countable conjunction universality theorem for one uniform formula. The present proof uses total exponentiation and internal binary coding in PA. Glazer’s announced fair-coin result reaches Q ; extending the full noise and Boolean classifications that far requires an alternative to the coding operations assumed here. A first target is an explicit elementary-arithmetic interface with all coding lemmas proved.
- Q2. Exact Borel complexity of cut fibers.** Classify the exact Borel or Wadge complexity of $\mathcal{E}_I$ from the order-theoretic position of I . The uniform Π_3^0 upper bound is proved, and cuts with a cofinal successor block have lower-complexity descriptions. Which remaining cuts attain the full bound? The closed-code description and the finite two-limit representation give concrete starting points for reductions.
- Q3. Automorphism-invariant laws.** Which cuts can be selected by a law invariant under the automorphism group of M ? Such a cut must be invariant, but this necessary condition need not provide biases whose local summability realizes it. The arbitrary-marker construction deliberately makes no invariance demand. Analyze orbit-constant biases first, then invariant dependent laws.
- Q4. Effective realization relative to a presentation.** Given a presentation of a countable model and descriptions of its target cuts, determine the oracle complexity needed to carry out the diagonal marker construction. The existence proof enumerates all true constraints $(S, a \in C_S)$ and is not an unconditional computable algorithm. An oracle-relative construction, with explicit moduli for the finite constraints, would connect the theorem to ProveIt’s computability work.
- Q5. Arithmetic genericity under biased laws.** Characterize the product measures that almost surely produce strongly CP-generic predicates. The exact coding cut alone cannot encode all the genericity requirements on definable tuples. Existing results rule out strongly CP-generic classes in nonstandard models [2]; thus a maximal coding cut already supplies an obstruction. Determine which additional divergence conditions on definable families are sufficient and necessary.
- Q6. Induction over an arbitrary modal predicate.** The exact cut formula allows an arbitrary modal pattern b , whereas the three-regime full-induction theorem assumes zero-modal noise. Classify higher induction in (M, P) when the modal expansion (M, b) has prescribed logical properties. Globally finite errors preserve induction with parameters, but infinite locally finite errors may have different effects when b cannot itself be defined from P .
- Q7. Boolean spectra with correlations inside each site.** Replace the product distribution across the input bits at a fixed site by an arbitrary finite joint law, while keeping the vectors at different sites independent. Determine the attainable Boolean cut tables. The minimal-support rule in Theorem 5.5 uses within-site independence, and XOR cancellation shows it can fail without it. Finite atom probabilities of these joint laws give a direct starting point.
- Q8. Quantitative approximate independence.** The pairwise proof only needs the mismatch variance to be negligible relative to the square of its mean along an exhaustion. Find useful uniform conditions on covariance errors that guarantee this for all internal codes simultaneously, and obtain optimal finite catalog bounds. The exact estimate (8.2) identifies the needed scale;

small pairwise errors individually may still accumulate too fast.

- Q9. More than one target under limited independence.** For pairwise-independent fair bits and a specified catalog $\mathcal{C} \subseteq \{0, 1\}^N$, compute the exact maximum of $\mathbb{P}(X \in \mathcal{C})$. The atom optimum is known, but the bound Kq_N may lose information about overlaps and geometry for multiword catalogs. Determine exact solutions for Hamming balls, linear catalogs, and catalogs of two or three words before approaching arbitrary sets.
- Q10. All-order sparse expansions and inversion.** Derive uniform higher-order expansions for $Q_N(K)$ and for the least budget attaining a prescribed confidence when $p_N \rightarrow 0$. The fixed Poisson parameter, the transition $Np_N \rightarrow \infty$, lattice effects, and a confidence approaching a staircase endpoint require different scales. The exact Hamming-layer formula supplies a natural starting point for asymptotic inversion and transseries methods.
- Q11. Algorithms for unequal biases.** The exact optimum ranks subsets by sums of costs $\log((1 - \varepsilon_i)/\varepsilon_i)$. Determine efficient certified algorithms for the first K subsets, especially with many repeated or structured biases. An arithmetic catalog imposed by a particular code enumeration can be much harder than an unrestricted optimal list; the computational resource model should make that distinction explicit.
- Q12. Growing or internally nonstandard alphabets.** Finite fixed alphabets admit straightforward modal-probability variants of the atom argument. The more interesting target is a uniform theory when alphabet sizes grow, or when the model regards the alphabet itself as nonstandard finite. How do finite-description costs, modal atoms, and the complexity of a single coding formula interact then?
- Q13. A verified end-to-end theorem.** Implement the arithmetic and probability interfaces of Section 11, culminating first in Theorem 8.1, then in Theorem 5.2. A meaningful deliverable would include a kernel-checked countable-model theorem and independent finite certificates for the source-coding and pairwise moment examples, with every unproved interface stated separately.
- Q14. Beyond countable models.** Determine replacements for the countable-null-union argument when the family of internal codes is uncountable, and compare the relevant ideals with those raised in Glazer’s seminar. Standard systems, cardinal invariants of null ideals, and local versus global measurability may provide useful conditional statements. The present article makes no claim about countably saturated uncountable models or their completed product measures.

The proposed problem is therefore solved at the level of arbitrary single-predicate product laws and countable independent Boolean spectra with biases uniformly bounded away from one, with an exact finite catalog theory and several sharp limitations. Its remaining questions concern symmetry, effective construction, weakened arithmetic, and dependence: restrictions under which the freedom to prescribe arbitrary cuts becomes a more rigid mathematical phenomenon.

13.1 Further questions and research

[write] **Added 6 October 2026, batch 114.** Following Vladimir Reshetnikov’s standing rule of 4 October 2026, every unproved claim of the source is carried here as an open question, with its source and what is missing. The fourteen questions above are the source’s own statements of open problems, each with its sketch; the write found nothing in the source to be wrong, so nothing is refuted. Status at the write:

- Question Q1: open. Glazer’s abstract announces the fair-coin case for Q ; no proof of it was read or is reproduced here, and his formula is not public. The proofs here use total exponentiation and internal coding in PA (Lemmas 2.2 and 6.1).
- Question Q2: open. Known here: the uniform Π_3^0 upper bound (Proposition 7.1) and Δ_3^0 for the cuts $\{x : x < a + n, n \text{ standard}\}$ (the remark following it). Missing: any lower bound or completeness result.
- Question Q3: open; nothing beyond the invariance of the selected cut is proved.
- Question Q4: open; the construction of Theorem 5.2 enumerates all true constraints (S, a) with $a \in C_S$, and no oracle bound is proved.
- Question Q5: open in general. Remark 13.1 below settles the laws whose almost-sure coding cut is M (the random predicate is almost surely not strongly CP-generic); for laws with a proper coding cut, the fair-coin law included, nothing is known here.
- Question Q6: open; Theorem 6.6 covers zero-modal laws ($p_t \leq 1/2$) only.
- Question Q7: open; the XOR example of Section 8 shows that the minimal-support rule of Theorem 5.5 fails without within-site independence.
- Question Q8: open; by (8.2) it suffices that $\text{Var}(Z_F)/m_F^2 \rightarrow 0$ along an exhaustion for every internal code, and no uniform condition guaranteeing this is proved.
- Question Q9: open; only the union bound $\min(1, Kq_N)$ of Theorem 8.2 is proved, and it is not claimed sharp for $K > 1$.
- Question Q10: open; Theorem 9.6 gives the leading-order limits only.
- Question Q11: open; Proposition 9.1 characterizes the optimal list, with no algorithm or complexity bound.
- Question Q12: open; only binary alphabets are treated.
- Question Q13: open. Section 11 is the source’s plan for it. At HEAD none of its modules exists; the declarations that do exist (the finite-model remark, beta coding of externally finite lists, the induction failure of a different formula) are listed in Section 1.3. Missing: every row of the table in Section 11, starting with the binary coding interface in raw PA models.
- Question Q14: open; the source makes no claim, and Glazer’s abstract records the obstruction for countably saturated models under the completed product measure.
- *Priority* (source: Sections 1 and 12, `SOURCE_AUDIT.txt`). The arbitrary-bias formula, the realization theorems, the Boolean spectra, the finite-marginal and weak-limit separations and the induction classification are proposed as new; the source found them in no source it inspected, and calls this “a bounded literature assessment, not a priority certificate”. Missing: a literature search beyond the cited sources; the write made none.

Remark 13.1 ([write] A maximal coding cut excludes strong CP-genericity, 6 October 2026). Let $M \models \text{PA}$ be nonstandard and let $P \subseteq M$ satisfy $\text{Cut}(P) = M$. Then P is a class of M in the sense of [2], and therefore, by Theorem 19 there, P is not strongly CP-generic. Consequently, if M is countable and nonstandard, then under every independent product law with $\text{Cut}(b) \cap S(\varepsilon) = M$ the random predicate is almost surely not strongly CP-generic. This holds, for example, for every zero-modal law with $W_x < \infty$ for all x (the first two regimes of Theorem 6.6) and for the law of Theorem 4.2 with $I = M$. This makes explicit the sentence of Question Q5 that a maximal coding cut already supplies an obstruction; it says nothing about laws with a proper coding cut.

Proof. Fix $a \in M$. Since $a \in \text{Cut}(P)$, there is $c < E(a)$ with $(M, P) \models \forall t < a (P(t) \leftrightarrow \text{Bit}(c, t))$. Hence

$$\{x \in P : x < a\} = \{x \in M : M \models x < a \wedge \text{Bit}(c, x)\},$$

and the right side is defined in M by an arithmetic formula with the parameters a and c : Bit is given by (2.1), in which E abbreviates the PA-definable function $t \mapsto 2^t$. So every bounded restriction of P is definable with parameters, which is the definition of a class in [2] (definability there means definability with parameters, and all models are models of PA). Their Theorem 19 says that in a nonstandard model no class is strongly CP-generic. For the probabilistic statement, Theorem 3.1 gives $\text{Cut}(P) = \text{Cut}(b) \cap S(\varepsilon) = M$ almost surely, and the first part applies to every predicate in that event. In the zero-modal case ($p_t \leq 1/2$ for all t) the modal predicate is empty, so $\text{Cut}(b) = M$ and $S(\varepsilon) = S(p) = \{x : W_x < \infty\}$, which is M when every W_x is finite. The law of Theorem 4.2 with $I = M$ has $\text{Cut}(P) = M$ almost surely by that theorem. $\square$

[write] **Independent check of the write, 6 October 2026.** An adversarial check made by the intake after the write (commit 90198e982), with its own code, found the write’s additions correct. It read Remark 13.1 against arXiv:2107.11867v3 (the conventions of its Section 1.1, the definition of a class in its Section 4, and Theorem 19 with its proof), found the paraphrases exact, and confirmed the definability argument and the probabilistic part (Theorem 3.1, the first two regimes of Theorem 6.6, and Theorem 4.2 with $I = M$). It confirmed, by `git grep` at the write, every declaration and file described under “Relation to the repository; formal status” in Section 1.3, and found the same-named copies in `Optimizations/Logic_Foundations_Optimizations.lean` that the first wording of that paragraph left out; the paragraph is corrected there with a dated note. It verified formula (12) of Proposition 25 of [5] against an exact linear program over exchangeable laws in 112 cases ($2 \leq n \leq 15$, eight biases), including the values $1/(n+1)$ for odd and $1/(n+2)$ for even n at $p = 1/2$; checked Theorem 9.4 and Proposition 9.1 by brute force over all words of length $N \leq 10$ (6,138 cases, exact rationals); and recomputed the convergence of Theorem 9.6 at $\lambda = 1$ for $N = 10^4$ and 10^5 , reproducing the printed values 0.45979872 and 0.73572209 for $K = N/4$ and $K = N$. No other item needed a correction.

References

- [1] Elliot Glazer. *Coin flipping on models of arithmetic to define the standard cut*. MOPA seminar, 17 October 2023. Official seminar abstract. <https://nylogic.github.io/mopa/2023/10/17/coin-flipping-on-models-of-arithmetic-to-define-the-standard-cut.html>. Accessed 5 October 2026.
- [2] Athar Abdul-Quader and James H. Schmerl. CP-generic expansions of models of Peano arithmetic. *Mathematical Logic Quarterly* **68** (2022), 171–177. doi:10.1002/malq.202100051. <https://arxiv.org/abs/2107.11867>. The inspected preprint was version 3; its class obstruction is Theorem 19.
- [3] Ioannis Kontoyiannis and Sergio Verdú. Optimal lossless data compression: Non-asymptotics and asymptotics. *IEEE Transactions on Information Theory* **60**(2) (2014), 777–795. doi:10.1109/TIT.2013.2291007. Preprint: *Lossless Data Compression at Finite Blocklengths*, <https://arxiv.org/abs/1212.2668>.
- [4] Shuqing Chen, Michelle Effros, and Victoria Kostina. Lossless source coding in the point-to-point, multiple access, and random access scenarios. *IEEE Transactions on Information Theory* **66**(11) (2020), 6688–6722. doi:10.1109/TIT.2020.3005155. <https://arxiv.org/abs/1902.03366>.
- [5] Itai Benjamini, Ori Gurel-Gurevich, and Ron Peled. *On K -wise Independent Distributions and Boolean Functions*. arXiv:1201.3261, uploaded 2012; the abstract identifies an earlier author-website manuscript. Proposition 25 gives the exact pairwise atom formula used here. <https://arxiv.org/abs/1201.3261>.

- [6] Endre Boros and András Prékopa. Closed form two-sided bounds for probabilities that at least r and exactly r out of n events occur. *Mathematics of Operations Research* **14**(2) (1989), 317–342. Bibliographic attribution and its application to pairwise independence were checked in [5].
- [7] Vladimir Reshetnikov, *ProveIt*, `Logic/PeanoArithmetic/ListCoding/README.md`. Repository file at the inspected revision. Describes the formula interfaces and their standard-model verification boundary. Accessed 5 October 2026.
- [8] Vladimir Reshetnikov, *ProveIt*, `Logic/PeanoArithmetic/NotFinitelyAxiomatizable/README.md`. Repository file at the inspected revision. Describes the raw-model, external finite coding, and definable-cut architecture. Accessed 5 October 2026.
- [9] Vladimir Reshetnikov, *ProveIt*, `PAFiniteBasisReduction/FiniteBetaCoding.lean`, under `Logic/PeanoArithmetic/NotFinitelyAxiomatizable/Lean/`. Inspected Lean source. External finite beta coding in raw PA models. Accessed 5 October 2026.
- [10] Vladimir Reshetnikov, *ProveIt*, `PAFiniteBasisReduction/EvaluatorCutContract.lean`, under `Logic/PeanoArithmetic/NotFinitelyAxiomatizable/Lean/`. Inspected Lean source. The evaluator-independent definable-cut contract and induction argument. Accessed 5 October 2026.